INFORME TÉCNICO

Periodo	2026-2
Actividad	Examen tercio 2
Fecha	2026/10/13
Código	1000098556
Nombre	Zharick Camila Chacón Gutiérrez

Informe técnico: solo será válido si en cada una de las capturas de pantalla se observa su código de alumno en el título o nombre de archivo. Para las capturas de pantalla puede utilizar la *Herramienta de Recortes* o ***Snipping Tool***, las teclas ***Win + PrtScn*** para guardar capturas de toda la ventana en la carpeta de imágenes, las teclas ***Win + G*** para obtener capturas desde la Game Bar, o las teclas ***Win + Shift + S*** para capturas instantáneas. Una vez finalizada la elaboración del informe técnico, este debe ser convertido a formato Adobe Acrobat .pdf y adjuntado al Quiz.

Contenido

1. P1 1
2. P2 11
3. P3 17
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7. P7 28

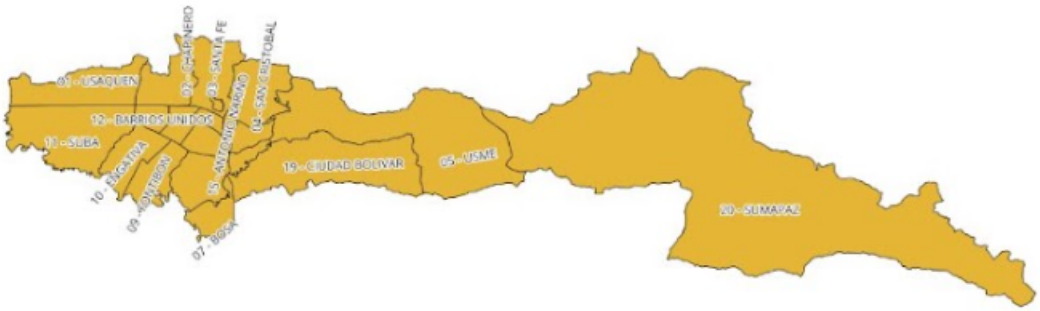
1. P1

Texto de la actividad:

1. A partir de los vectores, atributos y propiedades contenidas en la capa *Luminarias_Localidad.shp* que incluye el total de luminarias por localidad de la ciudad de Bogotá D.C. Colombia * 16 puntos

Procedimiento:

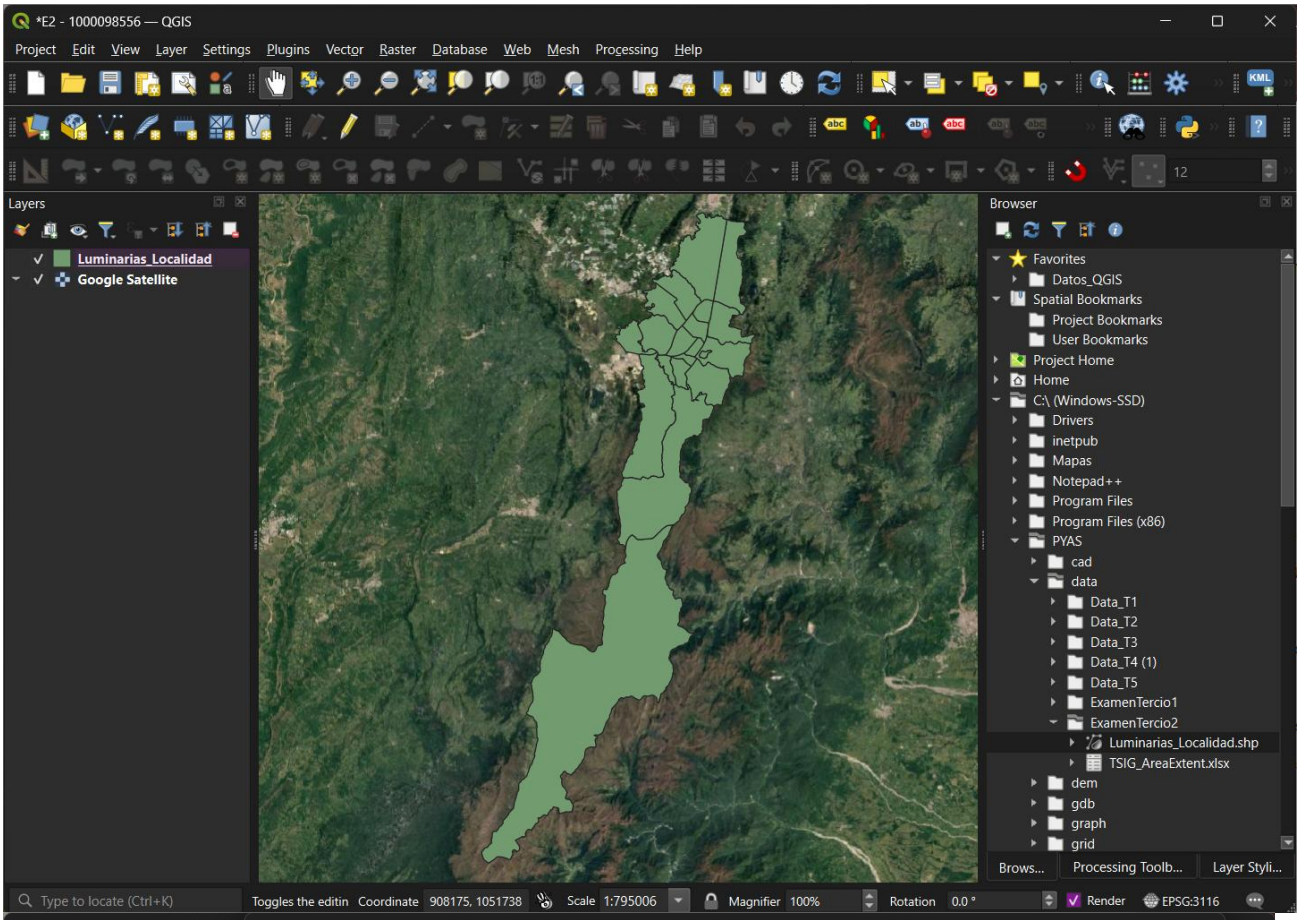
- Para resolver correctamente este examen, en QGIS desde el menú *Settings / General / Override System Locale*, establezca *User interface translation* como *American English* y *Locale (number, date and currency formats)* como *English United States (en_US)*
- Agregue la capa, asigne como *Data source encoding: Windows-1252,* consulte la tabla de atributos y las propiedades o metadatos
- Para contestar esta pregunta No es necesario realizar cálculos geométricos de campo o agregar campos en la tabla de atributos de la capa suministrada
- Para el cálculo del área del rectángulo envolvente utilice los valores de *Layer Properties / Information / Information from provider / Extent*, correspondientes a las coordenadas planas SW y NE que envuelven la capa, o utilice la herramienta *Layer Tools / Extract layer extent* para obtener el rectángulo envolvente
- Para los estadísticos utilice la herramienta *Vector Analysis / Statistics by categories*
- Esta pregunta solo será válida si consulta las propiedades solicitadas en QGIS

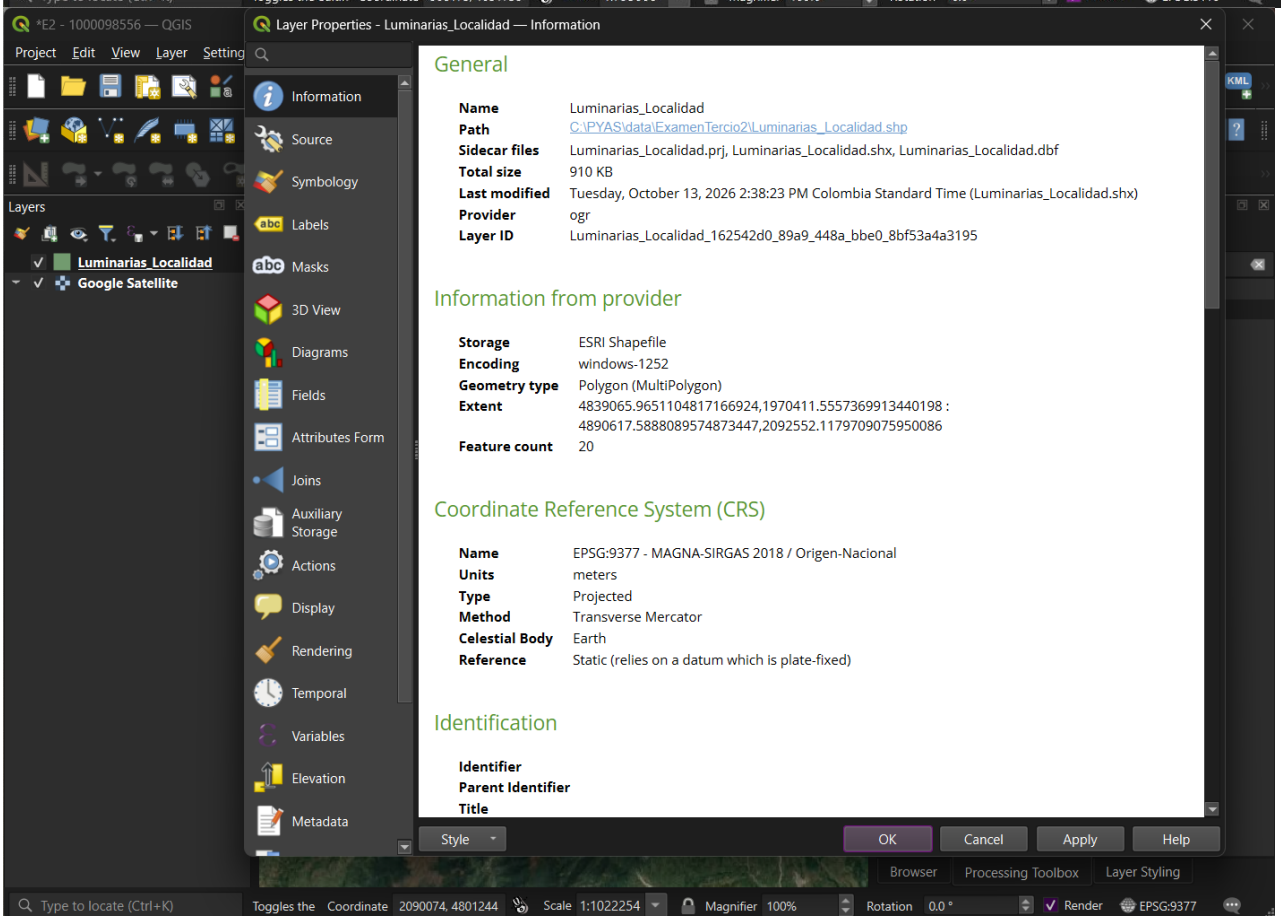


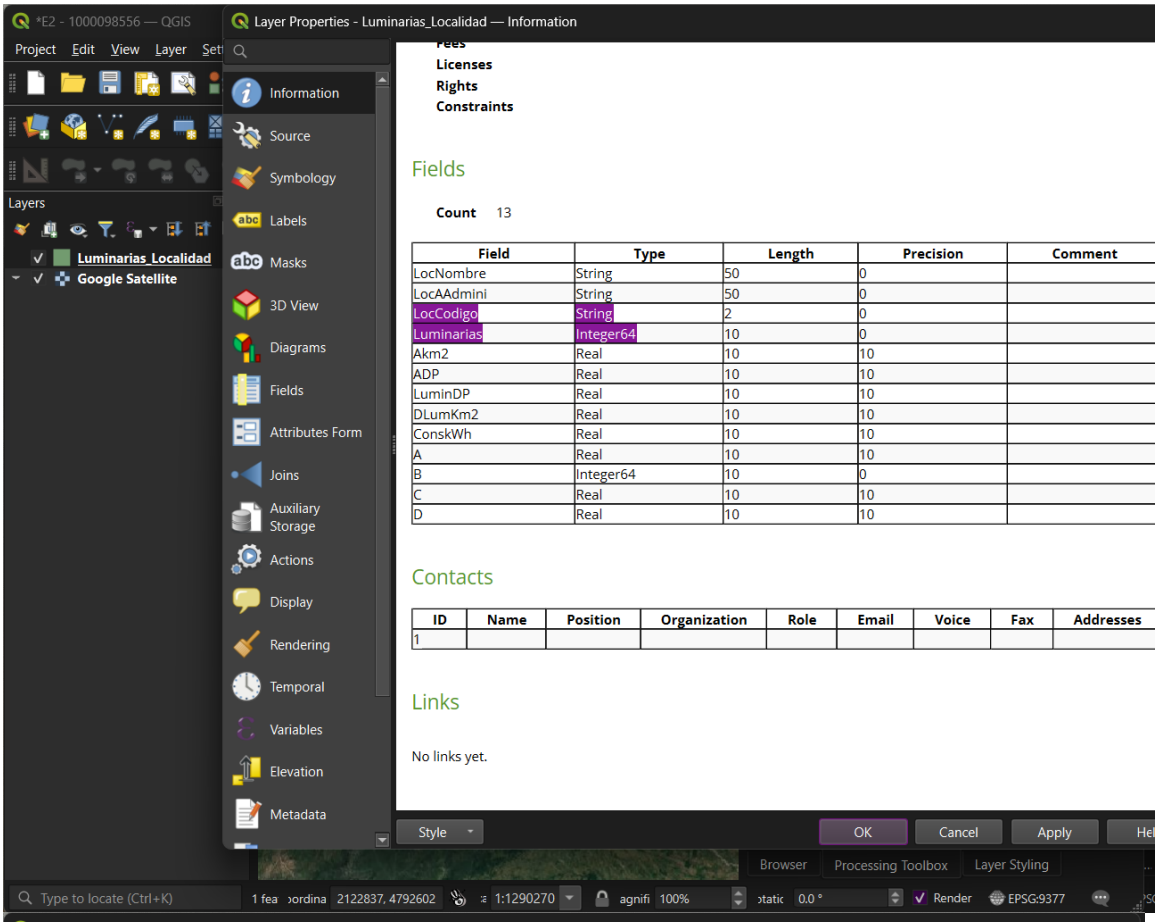
- [x] # de Localidades o registros en capa: 20
- [x] LocCodigo de la CANDELARIA: 17
- [x] CRS: 9377 - MAGNA-SIRGAS 2018 / Origen-Nacional
- [x] Tipo de campo LocCodigo: String
- [x] Tipo de campo Luminarias: Integer64
- [x] Límite este (m): 4890617.59
- [x] Límite oeste (m): 4839065.97
- [x] Límite norte (m): 2092552.12
- [x] Límite sur (m): 1970411.56
- [x] Área rectangular o de extent (km²): 6296.54
- [x] Localidades sin luminarias registradas: ANTONIO NARIÑO y SUMAPAZ
- [x] Total de luminarias Bogotá D.C.: 345786
- [x] Promedio (mean) de luminarias por Localidad: 19210.3
- [x] Luminarias en KENNEDY: 40209
- [x] Localidad con mayor número de luminarias: SUBA
- [x] Localidad con menor número de luminarias: CANDELARIA
- [ ] (No resuelto)

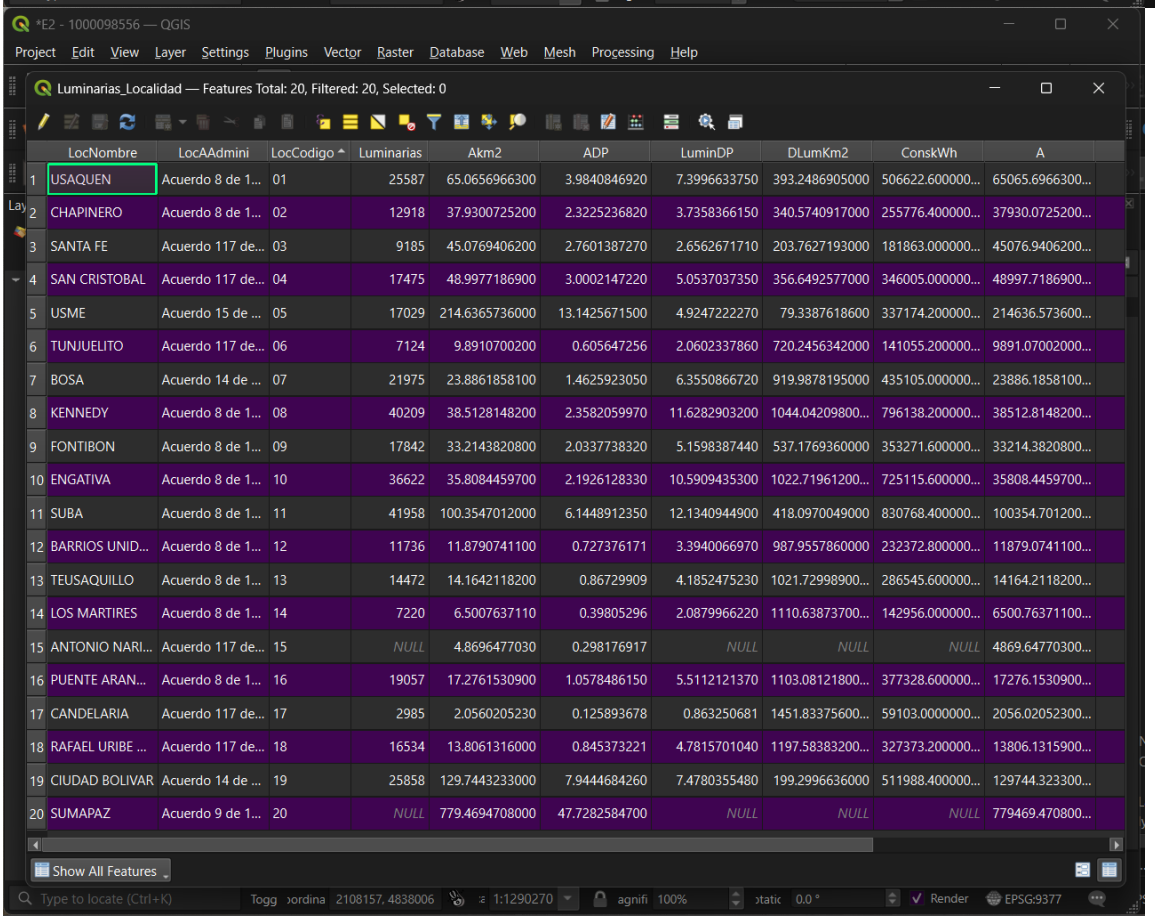
Capturas de pantalla:

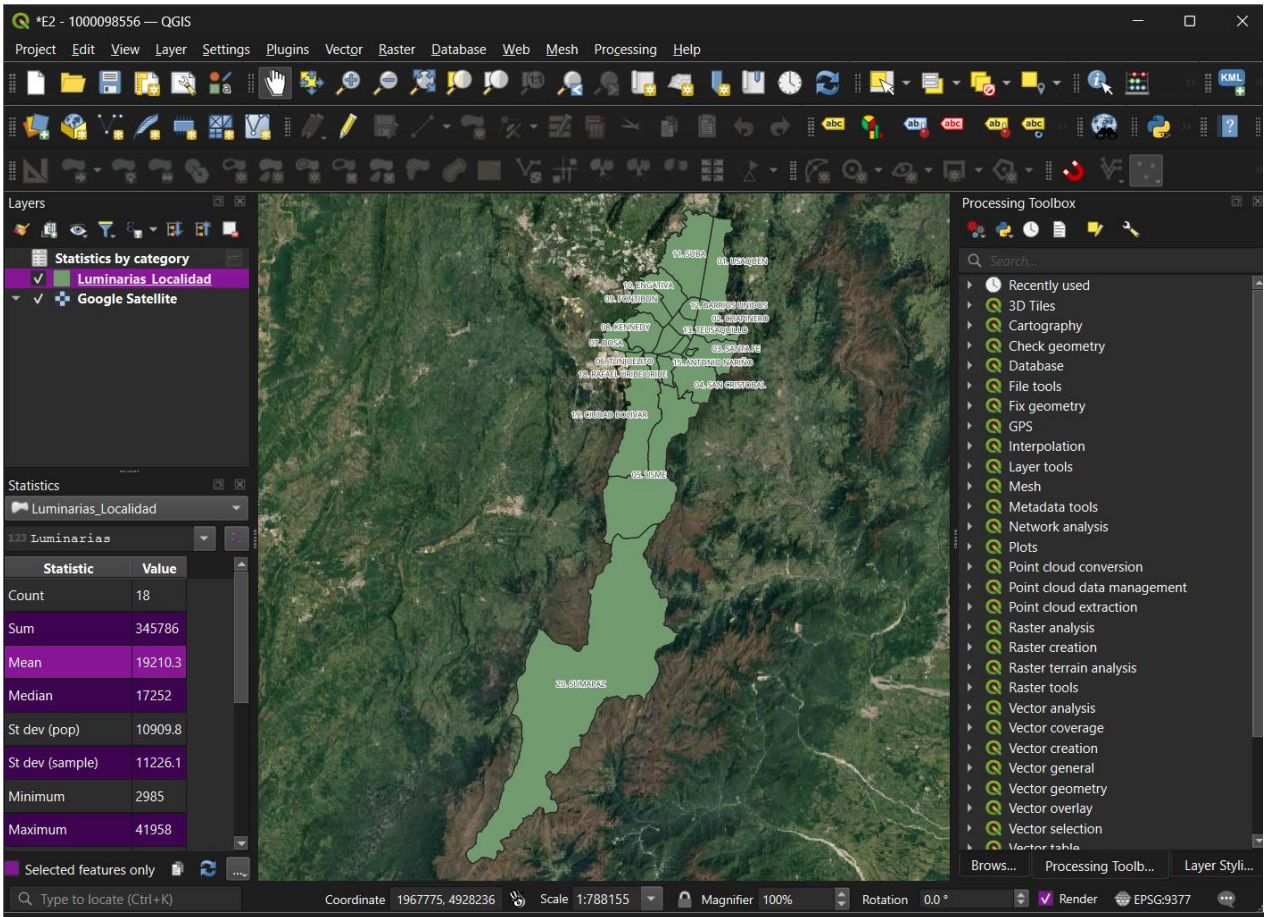
Capturas de pantalla:

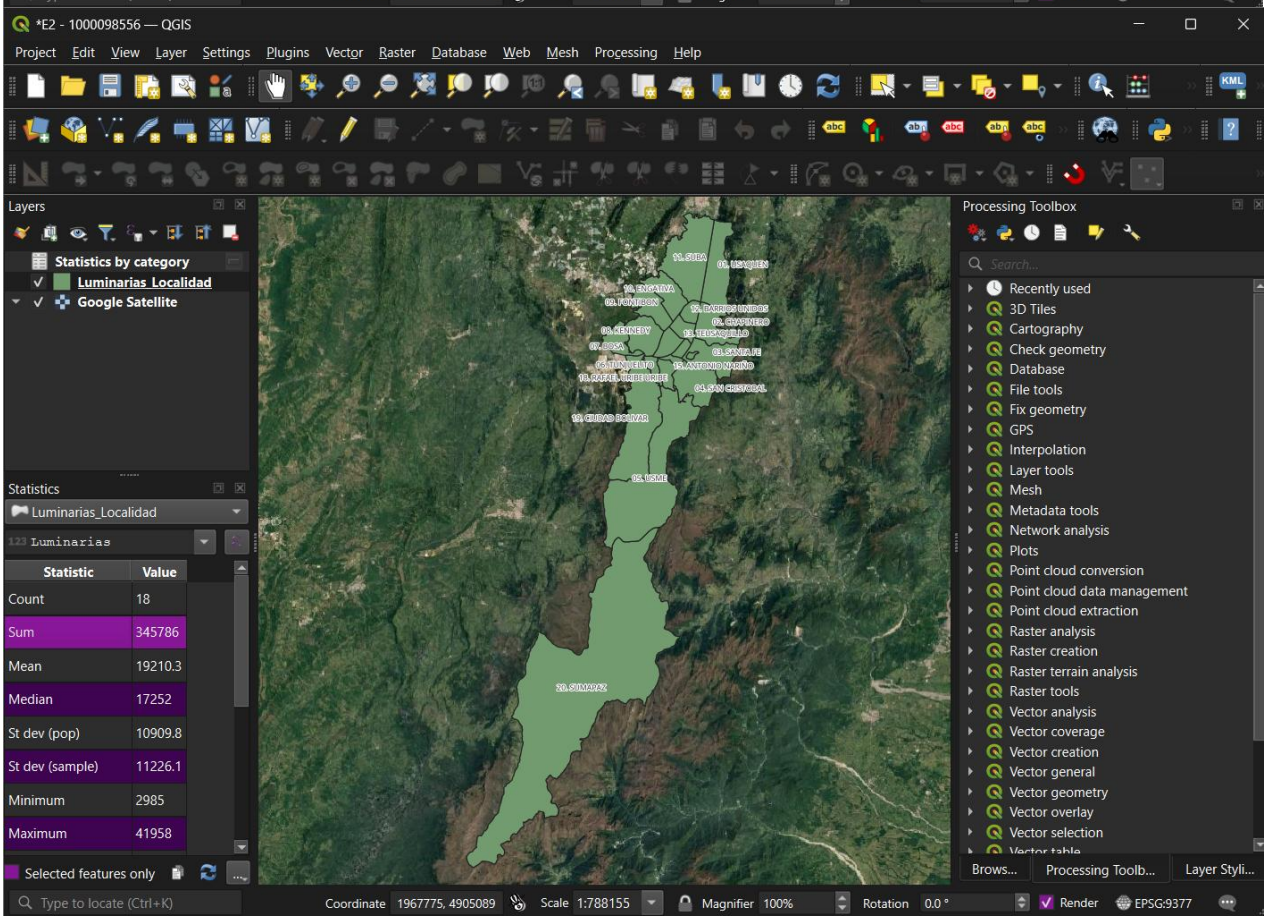

*E2 - 1000098556 — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Layers
Luminarias_Localidad
Google Satellite
Browser
Favorites
Datos_QGIS
Spatial Bookmarks
Project Bookmarks
User Bookmarks
Project Home
Home
C:\ (Windows-SSD)
Drivers
inetpub
Mapas
Notepad++
Program Files
Program Files (x86)
PYAS
cad
data
Data_T1
Data_T2
Data_T3
Data_T4 (1)
Data_T5
ExamenTercio1
ExamenTercio2
Luminarias_Localidad.shp
TSIG_AreaExtent.xlsx
dem
gdb
graph
grid
Brows...
Processing Toolb...
Layer Styli...
Type to locate (Ctrl+K)
Toggles the editin
Coordinate 908175, 1051738
Scale 1:795006
Magnifier 100%
Rotation 0.0 °
Render
EPSG:3116

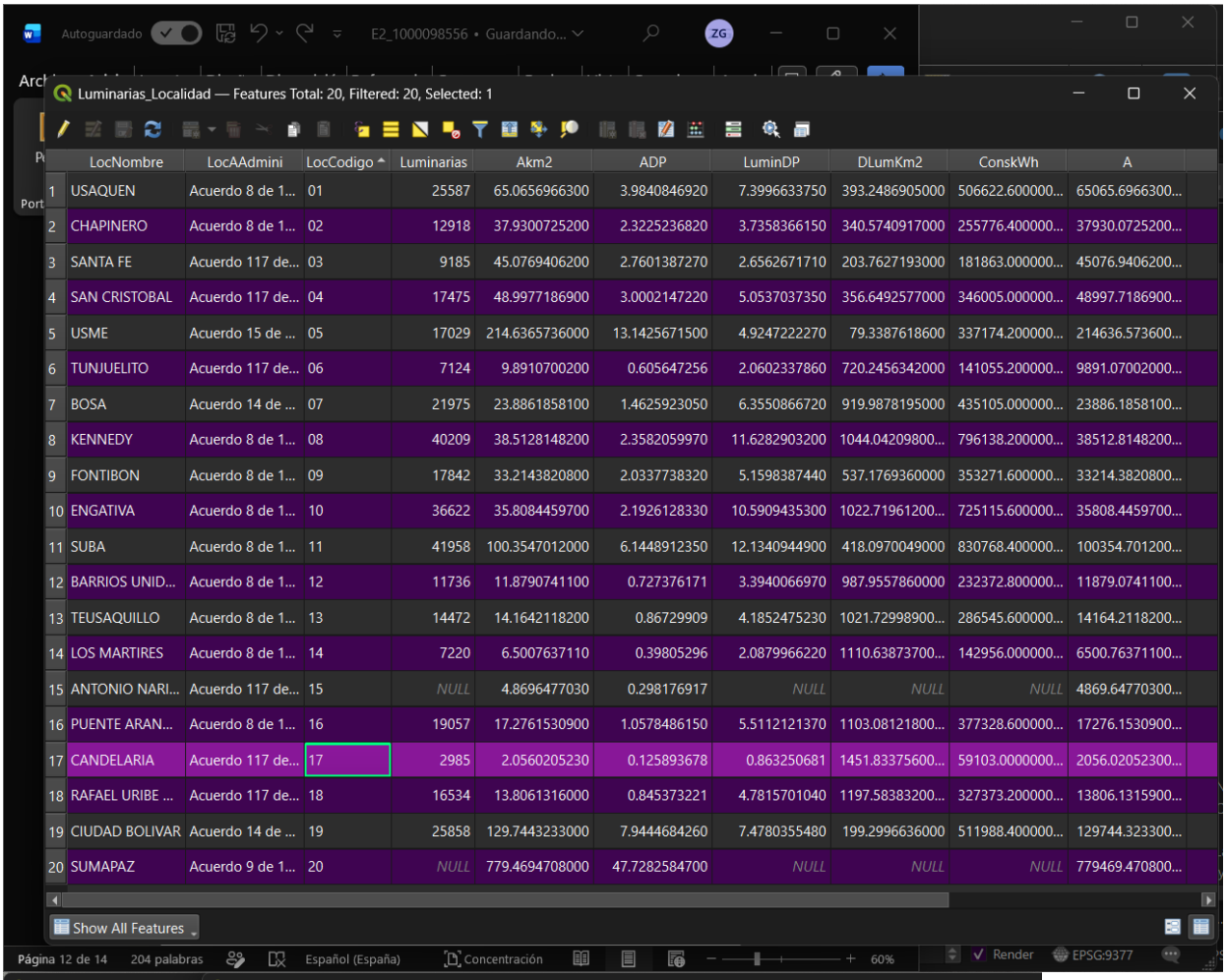

Layer Properties - Luminarias_Localidad — Information
Information
Source
Symbology
Labels
Masks
3D View
Diagrams
Fields
Attributes Form
Joins
Auxiliary Storage
Actions
Display
Rendering
Temporal
Variables
Elevation
Metadata
General
Name Luminarias_Localidad
Path C:\PYAS\data\ExamenTercio2\Luminarias_Localidad.shp
Sidecar files Luminarias_Localidad.prj, Luminarias_Localidad.shx, Luminarias_Localidad.dbf
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Last modified Tuesday, October 13, 2026 2:38:23 PM Colombia Standard Time (Luminarias_Localidad.shx)
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Information from provider
Storage ESRI Shapefile
Encoding windows-1252
Geometry type Polygon (MultiPolygon)
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Feature count 20
Coordinate Reference System (CRS)
Name EPSG:9377 - MAGNA-SIRGAS 2018 / Origen-Nacional
Units meters
Type Projected
Method Transverse Mercator
Celestial Body Earth
Reference Static (relies on a datum which is plate-fixed)
Identification
Identifier
Parent Identifier
Title
Style
OK
Cancel
Apply
Help
Browser
Processing Toolbox
Layer Styling
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Scale 1:1022254
Magnifier 100%
Rotation 0.0 °
Render
EPSG:9377


*E2 - 1000098556 — QGIS
Project Edit View Layer
Layers
Luminarias_Localidad
Google Satellite
Layer Properties - Luminarias_Localidad — Information
Information
Source
Symbology
Labels
Masks
3D View
Diagrams
Fields
Attributes Form
Joins
Auxiliary Storage
Actions
Display
Rendering
Temporal
Variables
Elevation
Metadata
Licenses
Rights
Constraints
Fields
Count 13
Field Type Length Precision Comment
LocNombre String 50 0
LocAAdmini String 50 0
LocCodigo String 2 0
Luminarias Integer64 10 0
Akm2 Real 10 10
ADP Real 10 10
LuminDP Real 10 10
DLumKm2 Real 10 10
ConskWh Real 10 10
A Real 10 10
B Integer64 10 0
C Real 10 10
D Real 10 10
Contacts
ID Name Position Organization Role Email Voice Fax Addresses
1
Links
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Style
OK
Cancel
Apply
Browser
Processing Toolbox
Layer Styling
Type to locate (Ctrl+K)
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100%
Render
EPSG:9377

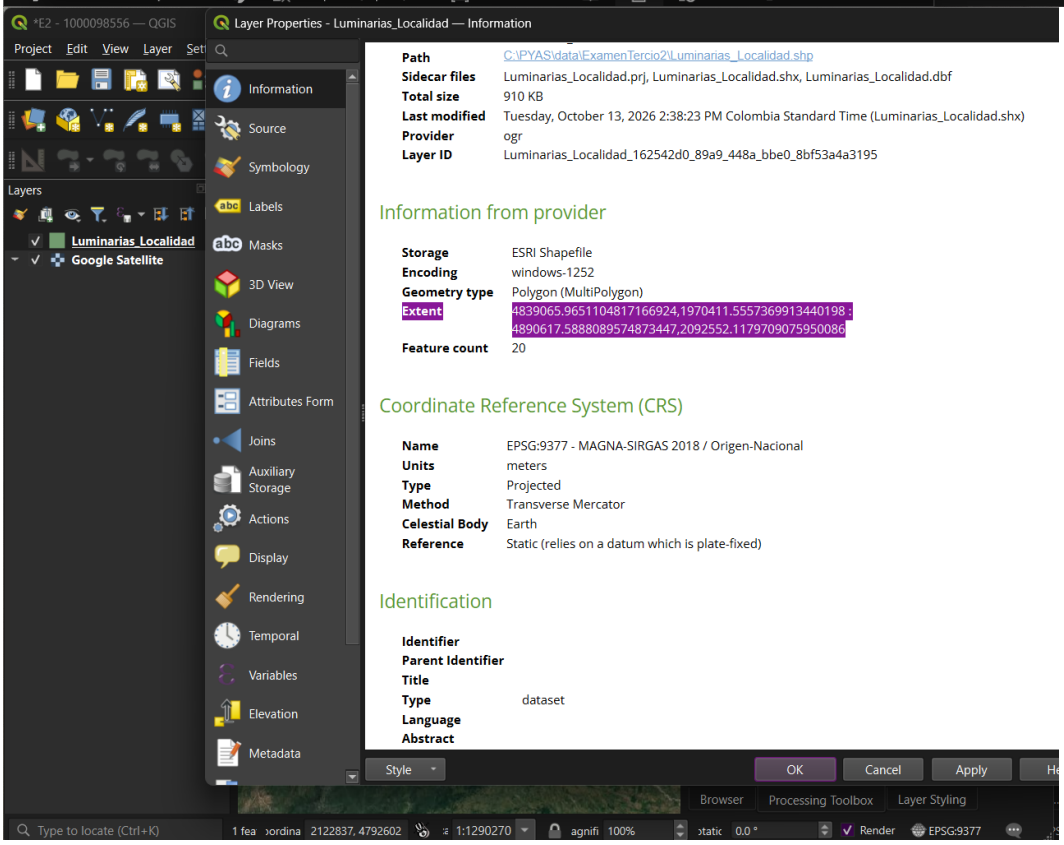

*E2 - 1000098556 — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Luminarias_Localidad — Features Total: 20, Filtered: 20, Selected: 0
LocNombre LocAAdmini LocCodigo Luminarias Akm2 ADP LuminDP DLumKm2 ConskWh A
1 USAQUEN Acuerdo 8 de 1... 01 25587 65.0656966300 3.9840846920 7.3996633750 393.2486905000 506622.600000... 65065.6966300...
2 CHAPINERO Acuerdo 8 de 1... 02 12918 37.9300725200 2.3225236820 3.7358366150 340.5740917000 255776.400000... 37930.0725200...
3 SANTA FE Acuerdo 117 de... 03 9185 45.0769406200 2.7601387270 2.6562671710 203.7627193000 181863.000000... 45076.9406200...
4 SAN CRISTOBAL Acuerdo 117 de... 04 17475 48.9977186900 3.0002147220 5.0537037350 356.6492577000 346005.000000... 48997.7186900...
5 USME Acuerdo 15 de ... 05 17029 214.6365736000 13.1425671500 4.9247222270 79.3387618600 337174.200000... 214636.573600...
6 TUNJUELITO Acuerdo 117 de... 06 7124 9.8910700200 0.605647256 2.0602337860 720.2456342000 141055.200000... 9891.07002000...
7 BOSA Acuerdo 14 de ... 07 21975 23.8861858100 1.4625923050 6.3550866720 919.9878195000 435105.000000... 23886.1858100...
8 KENNEDY Acuerdo 8 de 1... 08 40209 38.5128148200 2.3582059970 11.6282903200 1044.04209800... 796138.200000... 38512.8148200...
9 FONTIBON Acuerdo 8 de 1... 09 17842 33.2143820800 2.0337738320 5.1598387440 537.1769360000 353271.600000... 33214.3820800...
10 ENGATIVA Acuerdo 8 de 1... 10 36622 35.8084459700 2.1926128330 10.5909435300 1022.71961200... 725115.600000... 35808.4459700...
11 SUBA Acuerdo 8 de 1... 11 41958 100.3547012000 6.1448912350 12.1340944900 418.0970049000 830768.400000... 100354.701200...
12 BARRIOS UNID... Acuerdo 8 de 1... 12 11736 11.8790741100 0.727376171 3.3940066970 987.9557860000 232372.800000... 11879.0741100...
13 TEUSAQUILLO Acuerdo 8 de 1... 13 14472 14.1642118200 0.86729909 4.1852475230 1021.72998900... 286545.600000... 14164.2118200...
14 LOS MARTIRES Acuerdo 8 de 1... 14 7220 6.5007637110 0.39805296 2.0879966220 1110.63873700... 142956.000000... 6500.76371100...
15 ANTONIO NARI... Acuerdo 117 de... 15 NULL 4.8696477030 0.298176917 NULL NULL NULL 4869.64770300...
16 PUENTE ARAN... Acuerdo 8 de 1... 16 19057 17.2761530900 1.0578486150 5.5112121370 1103.08121800... 377328.600000... 17276.1530900...
17 CANDELARIA Acuerdo 117 de... 17 2985 2.0560205230 0.125893678 0.863250681 1451.83375600... 59103.0000000... 2056.02052300...
18 RAFAEL URIBE ... Acuerdo 117 de... 18 16534 13.8061316000 0.845373221 4.7815701040 1197.58383200... 327373.200000... 13806.1315900...
19 CIUDAD BOLIVAR Acuerdo 14 de ... 19 25858 129.7443233000 7.9444684260 7.4780355480 199.2996636000 511988.400000... 129744.323300...
20 SUMAPAZ Acuerdo 9 de 1... 20 NULL 779.4694708000 47.7282584700 NULL NULL NULL 779469.470800...
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Type to locate (Ctrl+K)
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100%
Render
EPSG:9377

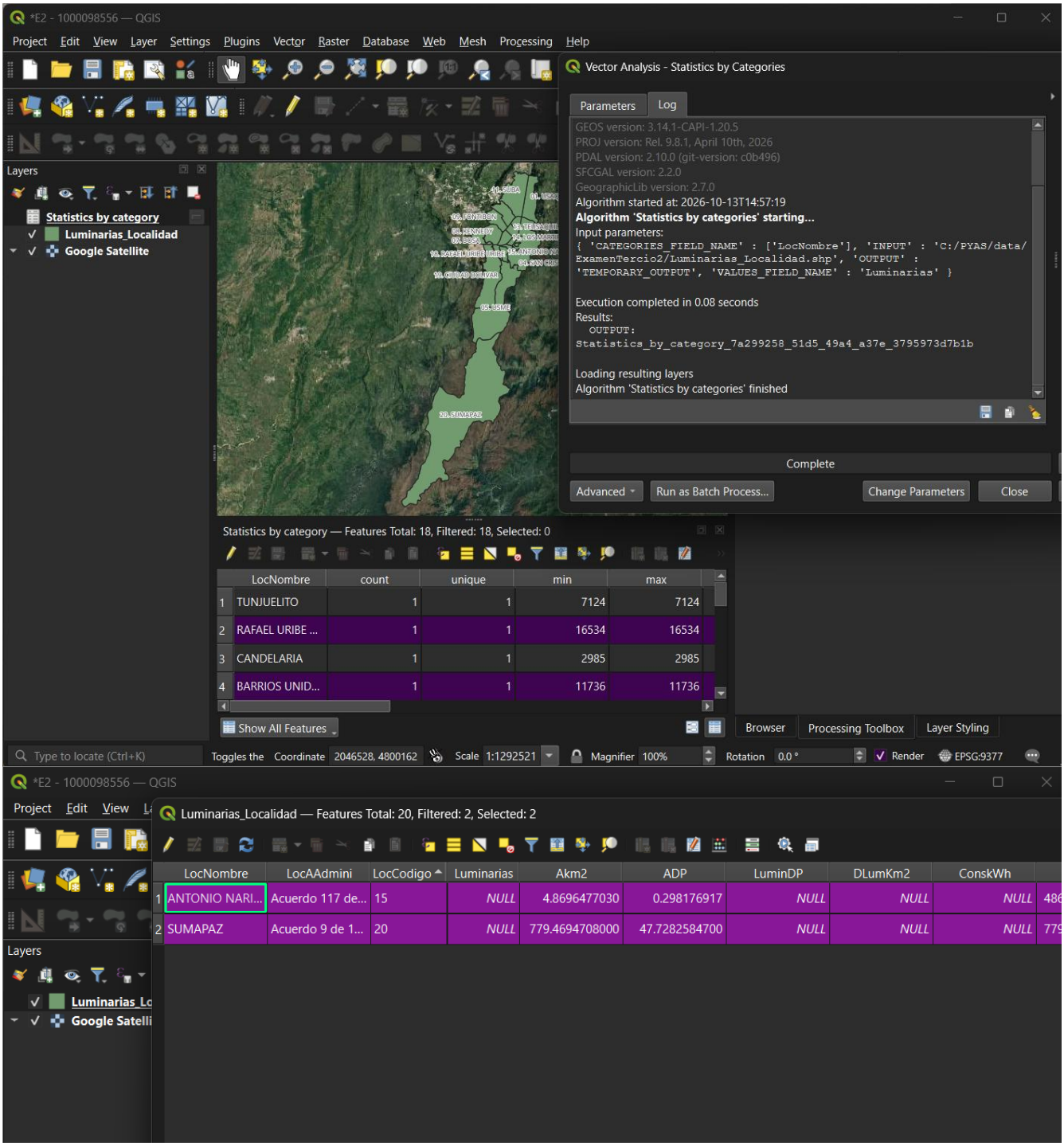

*E2 - 1000098556 — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Layers
Statistics by category
Luminarias_Localidad
Google Satellite
Statistics
Luminarias_Localidad
Luminarias
Statistic Value
Count 18
Sum 345786
Mean 19210.3
Median 17252
St dev (pop) 10909.8
St dev (sample) 11226.1
Minimum 2985
Maximum 41958
Selected features only
Type to locate (Ctrl+K)
Processing Toolbox
Recently used
3D Tiles
Cartography
Check geometry
Database
File tools
Fix geometry
GPS
Interpolation
Layer tools
Mesh
Metadata tools
Network analysis
Plots
Point cloud conversion
Point cloud data management
Point cloud extraction
Raster analysis
Raster creation
Raster terrain analysis
Raster tools
Vector analysis
Vector coverage
Vector creation
Vector general
Vector geometry
Vector overlay
Vector selection
Brows...
Processing Toolb...
Layer Styli...
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Magnifier 100%
Rotation 0.0 °
Render
EPSG:9377

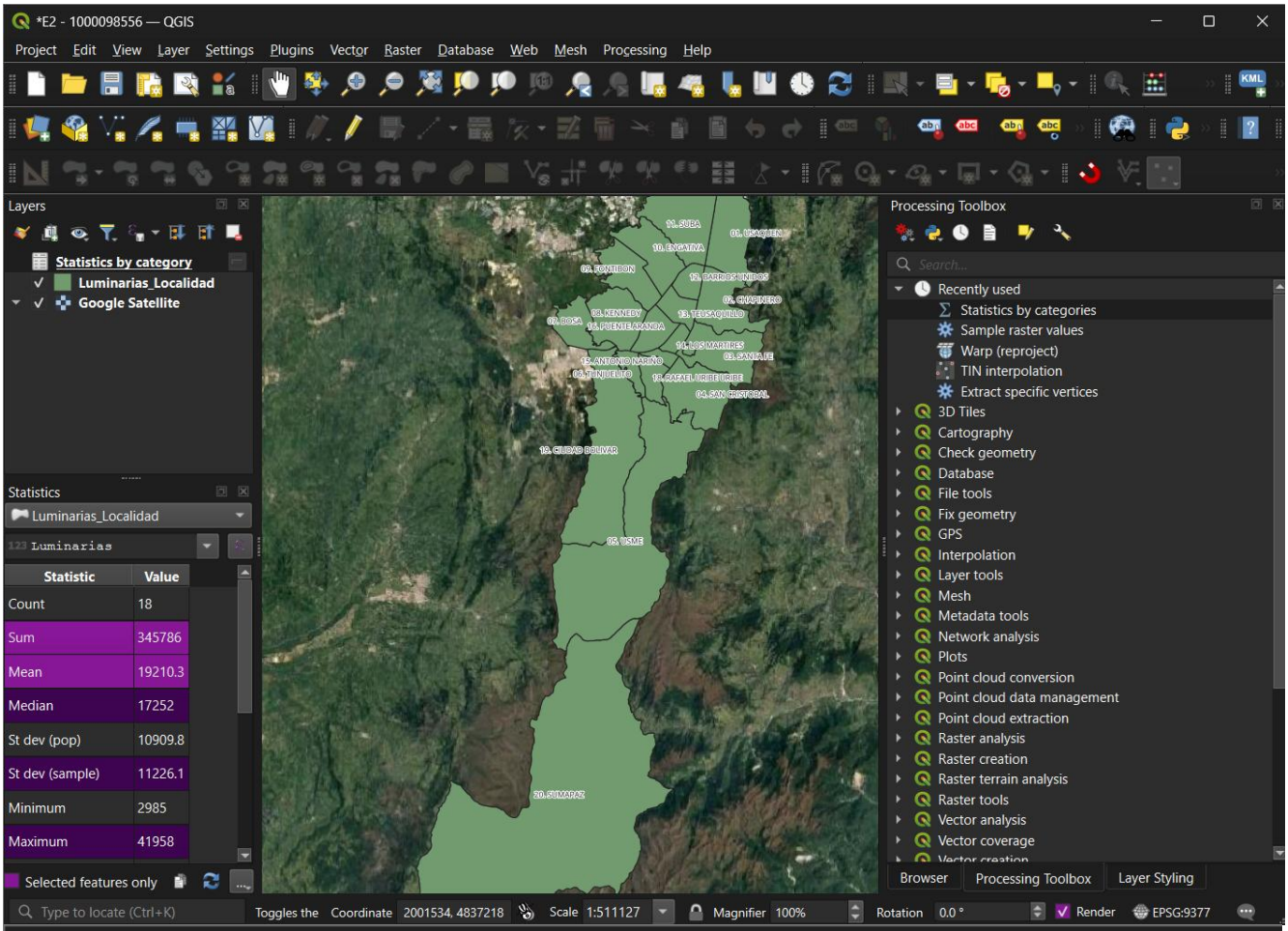

*E2 - 1000098556 — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Layers
Statistics by category
Luminarias_Localidad
Google Satellite
Statistics
Luminarias_Localidad
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St dev (pop) 10909.8
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Point cloud conversion
Point cloud data management
Point cloud extraction
Raster analysis
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Raster terrain analysis
Raster tools
Vector analysis
Vector coverage
Vector creation
Vector general
Vector geometry
Vector overlay
Vector selection
Brows...
Processing Toolb...
Layer Styli...
Coordinate 1967775, 4905089
Scale 1:788155
Magnifier 100%
Rotation 0.0 °
Render
EPSG:9377

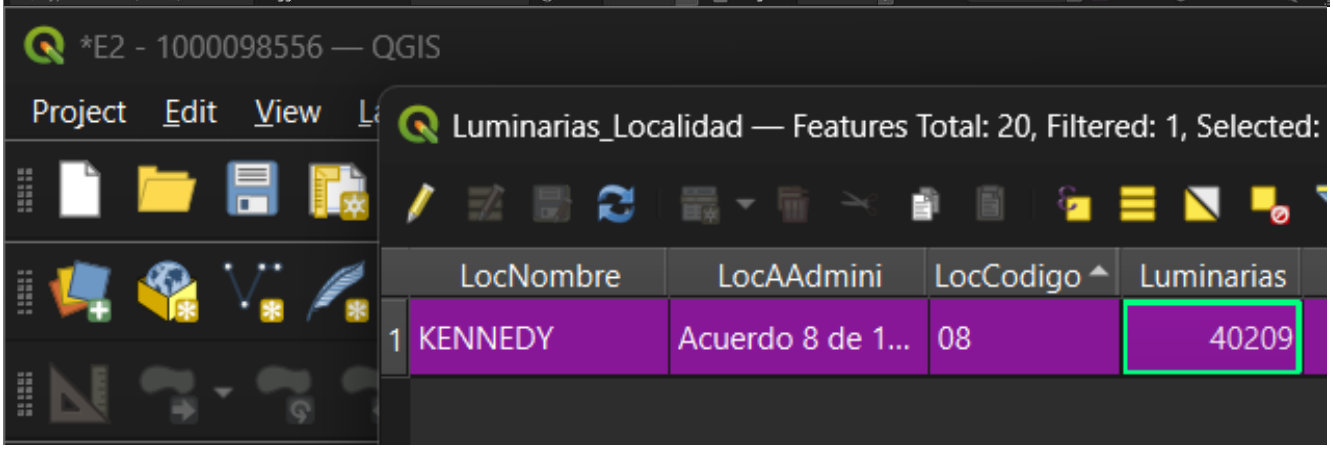


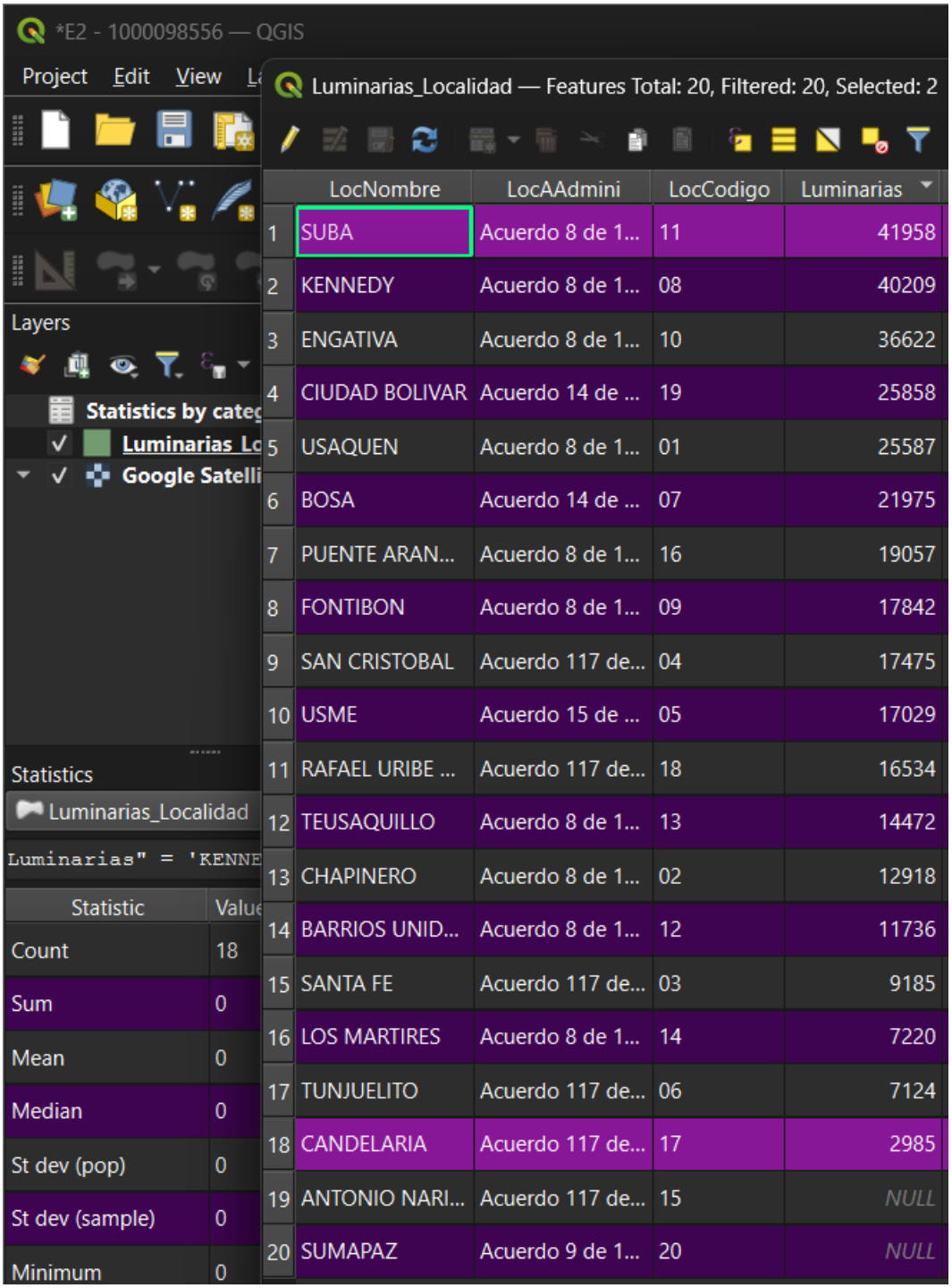
	LocNombre	LocAAdmini	LocCodigo	Luminarias	Akm2	ADP	LuminDP	DLumKm2	ConskWh	A
1	USAQUEN	Acuerdo 8 de 1...	01	25587	65.0656966300	3.9840846920	7.3996633750	393.2486905000	506622.600000...	65065.6966300...
2	CHAPINERO	Acuerdo 8 de 1...	02	12918	37.9300725200	2.3225236820	3.7358366150	340.5740917000	255776.400000...	37930.0725200...
3	SANTA FE	Acuerdo 117 de...	03	9185	45.0769406200	2.7601387270	2.6562671710	203.7627193000	181863.000000...	45076.9406200...
4	SAN CRISTOBAL	Acuerdo 117 de...	04	17475	48.9977186900	3.0002147220	5.0537037350	356.6492577000	346005.000000...	48997.7186900...
5	USME	Acuerdo 15 de ...	05	17029	214.6365736000	13.1425671500	4.9247222270	79.3387618600	337174.200000...	214636.573600...
6	TUNJUELITO	Acuerdo 117 de...	06	7124	9.8910700200	0.605647256	2.0602337860	720.2456342000	141055.200000...	9891.07002000...
7	BOSA	Acuerdo 14 de ...	07	21975	23.8861858100	1.4625923050	6.3550866720	919.9878195000	435105.000000...	23886.1858100...
8	KENNEDY	Acuerdo 8 de 1...	08	40209	38.5128148200	2.3582059970	11.6282903200	1044.04209800...	796138.200000...	38512.8148200...
9	FONTIBON	Acuerdo 8 de 1...	09	17842	33.2143820800	2.0337738320	5.1598387440	537.1769360000	353271.600000...	33214.3820800...
10	ENGATIVA	Acuerdo 8 de 1...	10	36622	35.8084459700	2.1926128330	10.5909435300	1022.71961200...	725115.600000...	35808.4459700...
11	SUBA	Acuerdo 8 de 1...	11	41958	100.3547012000	6.1448912350	12.1340944900	418.0970049000	830768.400000...	100354.701200...
12	BARRIOS UNID...	Acuerdo 8 de 1...	12	11736	11.8790741100	0.727376171	3.3940066970	987.9557860000	232372.800000...	11879.0741100...
13	TEUSAQUILLO	Acuerdo 8 de 1...	13	14472	14.1642118200	0.86729909	4.1852475230	1021.72998900...	286545.600000...	14164.2118200...
14	LOS MARTIRES	Acuerdo 8 de 1...	14	7220	6.5007637110	0.39805296	2.0879966220	1110.63873700...	142956.000000...	6500.76371100...
15	ANTONIO NARI...	Acuerdo 117 de...	15	NULL	4.8696477030	0.298176917	NULL	NULL	NULL	4869.64770300...
16	PUENTE ARAN...	Acuerdo 8 de 1...	16	19057	17.2761530900	1.0578486150	5.5112121370	1103.08121800...	377328.600000...	17276.1530900...
17	CANDELARIA	Acuerdo 117 de...	17	2985	2.0560205230	0.125893678	0.863250681	1451.83375600...	59103.0000000...	2056.02052300...
18	RAFAEL URIBE ...	Acuerdo 117 de...	18	16534	13.8061316000	0.845373221	4.7815701040	1197.58383200...	327373.200000...	13806.1315900...
19	CIUDAD BOLIVAR	Acuerdo 14 de ...	19	25858	129.7443233000	7.9444684260	7.4780355480	199.2996636000	511988.400000...	129744.323300...
20	SUMAPAZ	Acuerdo 9 de 1...	20	NULL	779.4694708000	47.7282584700	NULL	NULL	NULL	779469.470800...



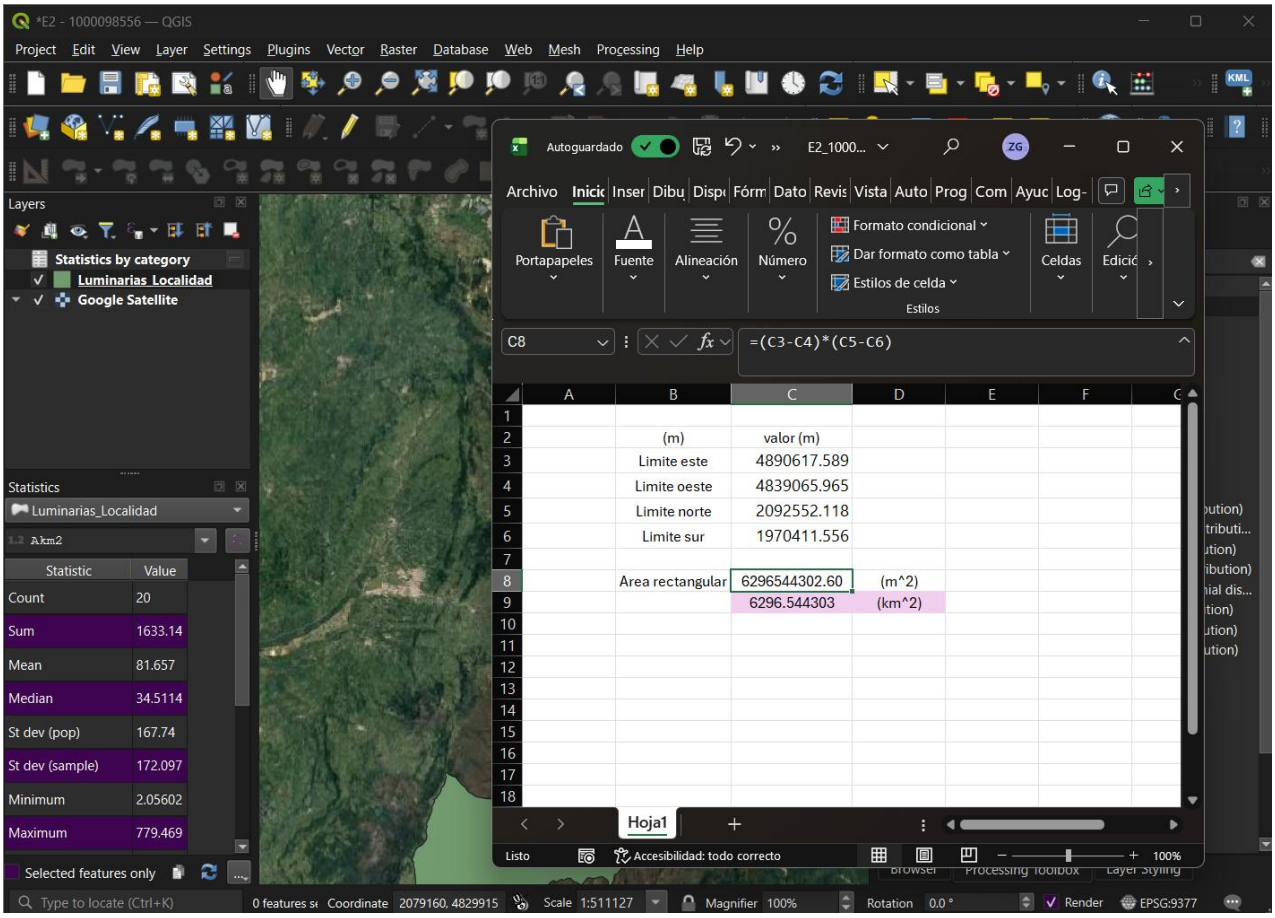

*E2 - 1000098556 — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Vector Analysis - Statistics by Categories
Parameters
Log
GEOS version: 3.14.1-CAPI-1.20.5
PROJ version: Rel. 9.8.1, April 10th, 2026
PDAL version: 2.10.0 (git-version: c0b496)
SFCGAL version: 2.2.0
GeographicLib version: 2.7.0
Algorithm started at: 2026-10-13T14:57:19
Algorithm 'Statistics by categories' starting...
Input parameters:
{ 'CATEGORIES_FIELD_NAME' : ['LocNombre'], 'INPUT' : 'C:/PYAS/data/ExamenTercio2/Luminarias_Localidad.shp', 'OUTPUT' : 'TEMPORARY_OUTPUT', 'VALUES_FIELD_NAME' : 'Luminarias' }
Execution completed in 0.08 seconds
Results:
OUTPUT:
Statistics_by_category_7a299258_51d5_49a4_a37e_3795973d7b1b
Loading resulting layers
Algorithm 'Statistics by categories' finished
Complete
Advanced
Run as Batch Process...
Change Parameters
Close
Layers
Statistics by category
Luminarias_Localidad
Google Satellite
Statistics by category — Features Total: 18, Filtered: 18, Selected: 0
LocNombre count unique min max
1 TUNJUELITO 1 1 7124 7124
2 RAFAEL URIBE ... 1 1 16534 16534
3 CANDELARIA 1 1 2985 2985
4 BARRIOS UNID... 1 1 11736 11736
Show All Features
Browser
Processing Toolbox
Layer Styling
Type to locate (Ctrl+K)
Toggles the Coordinate 2046528, 4800162
Scale 1:1292521
Magnifier 100%
Rotation 0.0 °
Render
EPSG:9377
*E2 - 1000098556 — QGIS
Project Edit View
Luminarias_Localidad — Features Total: 20, Filtered: 2, Selected: 2
LocNombre LocAAdmini LocCodigo Luminarias Akm2 ADP LuminDP DLumKm2 ConskWh
1 ANTONIO NARI... Acuerdo 117 de... 15 NULL 4.8696477030 0.298176917 NULL NULL NULL
2 SUMAPAZ Acuerdo 9 de 1... 20 NULL 779.4694708000 47.7282584700 NULL NULL NULL
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Luminarias_Lo
Google Satelli


*E2 - 1000098556 — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Layers
Statistics by category
Luminarias_Localidad
Google Satellite
Statistics
Luminarias_Localidad
Luminarias
Statistic Value
Count 18
Sum 345786
Mean 19210.3
Median 17252
St dev (pop) 10909.8
St dev (sample) 11226.1
Minimum 2985
Maximum 41958
Selected features only
Processing Toolbox
Recently used
Statistics by categories
Sample raster values
Warp (reproject)
TIN interpolation
Extract specific vertices
3D Tiles
Cartography
Check geometry
Database
File tools
Fix geometry
GPS
Interpolation
Layer tools
Mesh
Metadata tools
Network analysis
Plots
Point cloud conversion
Point cloud data management
Point cloud extraction
Raster analysis
Raster creation
Raster terrain analysis
Raster tools
Vector analysis
Vector coverage
Browser
Processing Toolbox
Layer Styling
Type to locate (Ctrl+K)
Toggles the
Coordinate 2001534, 4837218
Scale 1:511127
Magnifier 100%
Rotation 0.0 °
Render
EPSG:9377


*E2 - 1000098556 — QGIS
Project Edit View
Luminarias_Localidad — Features Total: 20, Filtered: 1, Selected:
LocNombre LocAAdmini LocCodigo Luminarias
1 KENNEDY Acuerdo 8 de 1... 08 40209



	LocNombre	LocAAdmini	LocCodigo	Luminarias
1	SUBA	Acuerdo 8 de 1...	11	41958
2	KENNEDY	Acuerdo 8 de 1...	08	40209
3	ENGATIVA	Acuerdo 8 de 1...	10	36622
4	CIUDAD BOLIVAR	Acuerdo 14 de ...	19	25858
5	USAQUEN	Acuerdo 8 de 1...	01	25587
6	BOSA	Acuerdo 14 de ...	07	21975
7	PUENTE ARAN...	Acuerdo 8 de 1...	16	19057
8	FONTIBON	Acuerdo 8 de 1...	09	17842
9	SAN CRISTOBAL	Acuerdo 117 de...	04	17475
10	USME	Acuerdo 15 de ...	05	17029
11	RAFAEL URIBE ...	Acuerdo 117 de...	18	16534
12	TEUSAQUILLO	Acuerdo 8 de 1...	13	14472
13	CHAPINERO	Acuerdo 8 de 1...	02	12918
14	BARRIOS UNID...	Acuerdo 8 de 1...	12	11736
15	SANTA FE	Acuerdo 117 de...	03	9185
16	LOS MARTIRES	Acuerdo 8 de 1...	14	7220
17	TUNJUELITO	Acuerdo 117 de...	06	7124
18	CANDELARIA	Acuerdo 117 de...	17	2985
19	ANTONIO NARI...	Acuerdo 117 de...	15	NULL
20	SUMAPAZ	Acuerdo 9 de 1...	20	NULL



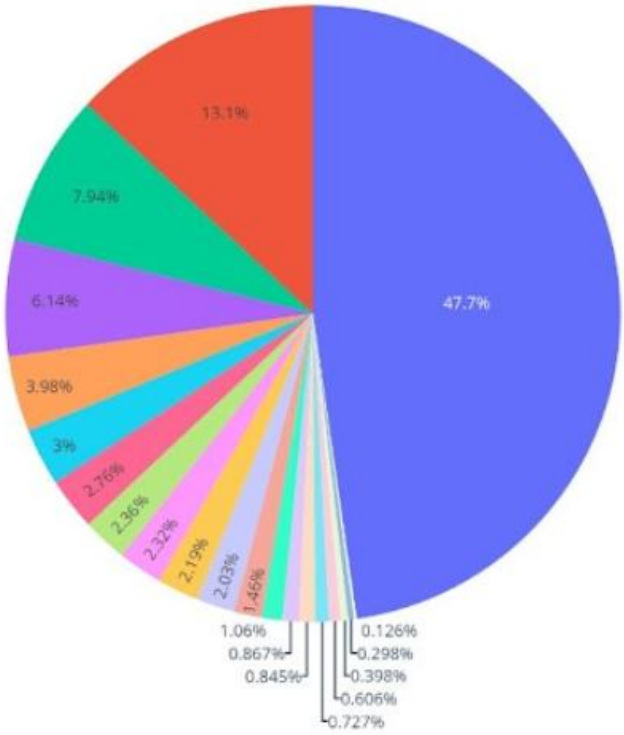
2. P2

Texto de la actividad:

2. Distribución porcentual de Área: cree los campos de atributos *Akm2* y *ADP* real o double (10 posiciones enteras y 10 decimales de precisión), calcule el área planar en km² de cada localidad y su distribución porcentual * 6 puntos

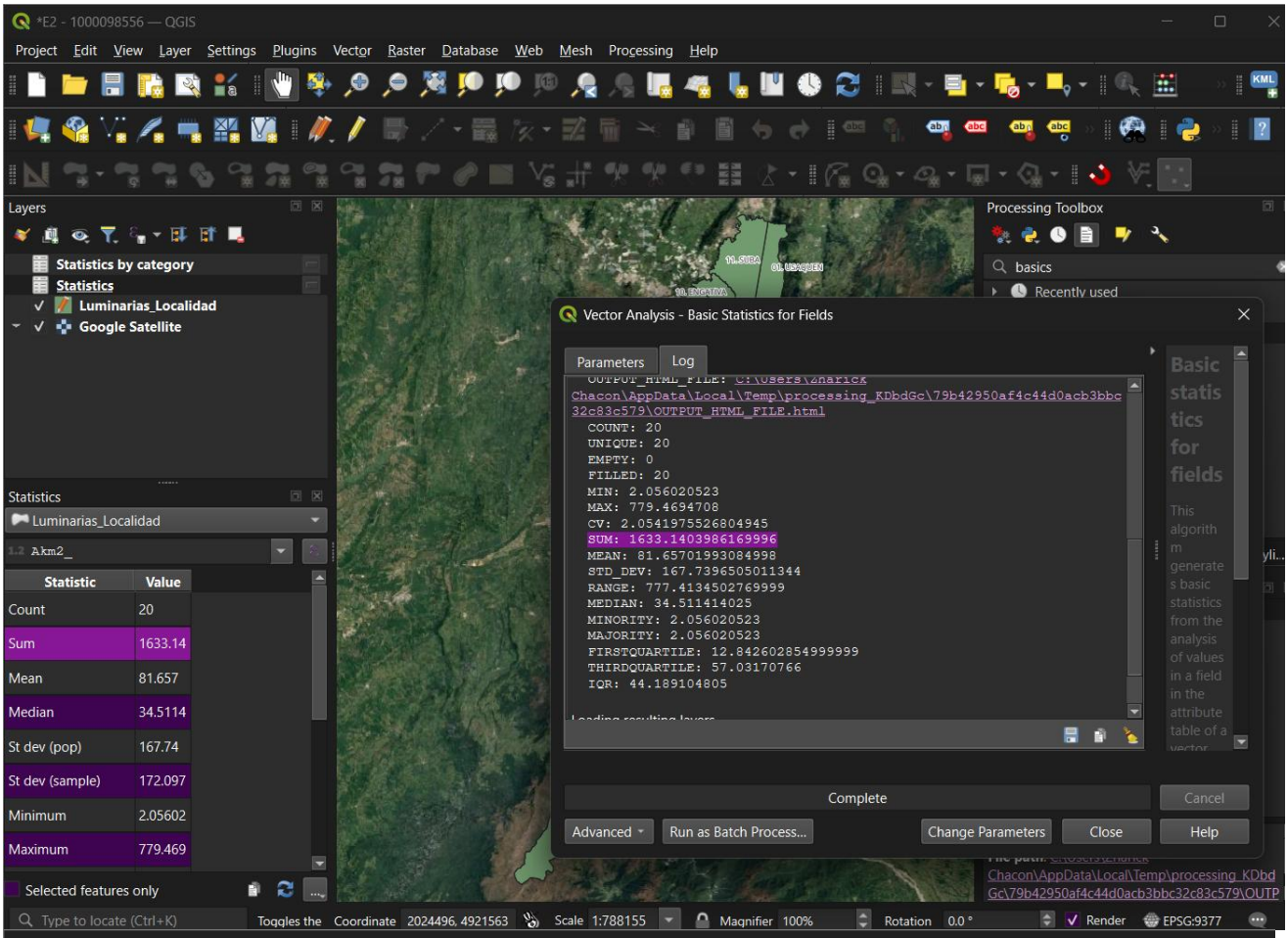
Procedimiento:

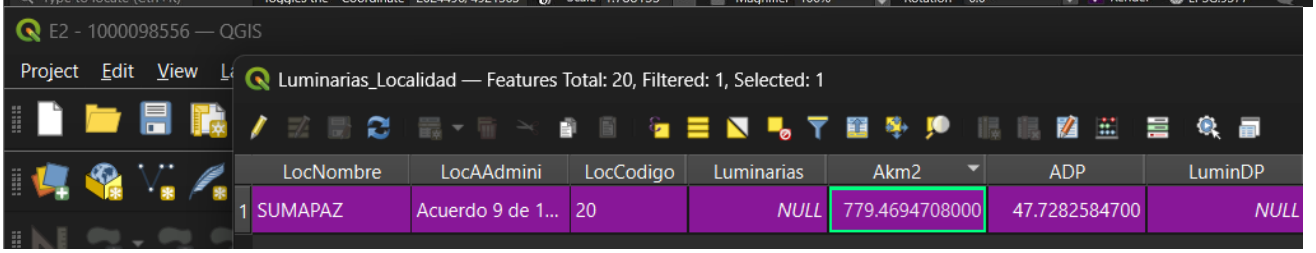
- Desde el *Field Calculator* de la tabla de atributos realice el cálculo solicitado de área. *Akm2* = area(@geometry)/1000000
- Con la herramienta *Vector Analysis / Basic statistics for fields*, obtenga la sumatoria exacta del campo *Akm2*
- Desde el *Field Calculator* de la tabla de atributos realice el cálculo solicitado de distribución porcentual de área. ADP = ("Akm2" /SumatoriaArea)*100
- Tenga en cuenta que debe reemplazar *SumatoriaArea* por el valor de la sumatoria obtenida
- Verifique que la sumatoria de las distribuciones porcentuales sea 100%
- Utilizando el complemento *DataPlotly, Advanced Charts* o la herramienta *Processing Toolbox / Plots / Bar Plot*, cree una gráfica de pastel o de barras representando las distribuciones porcentuales obtenidas por localidad
- Esta pregunta solo será válida si realiza los cálculos solicitados en QGIS

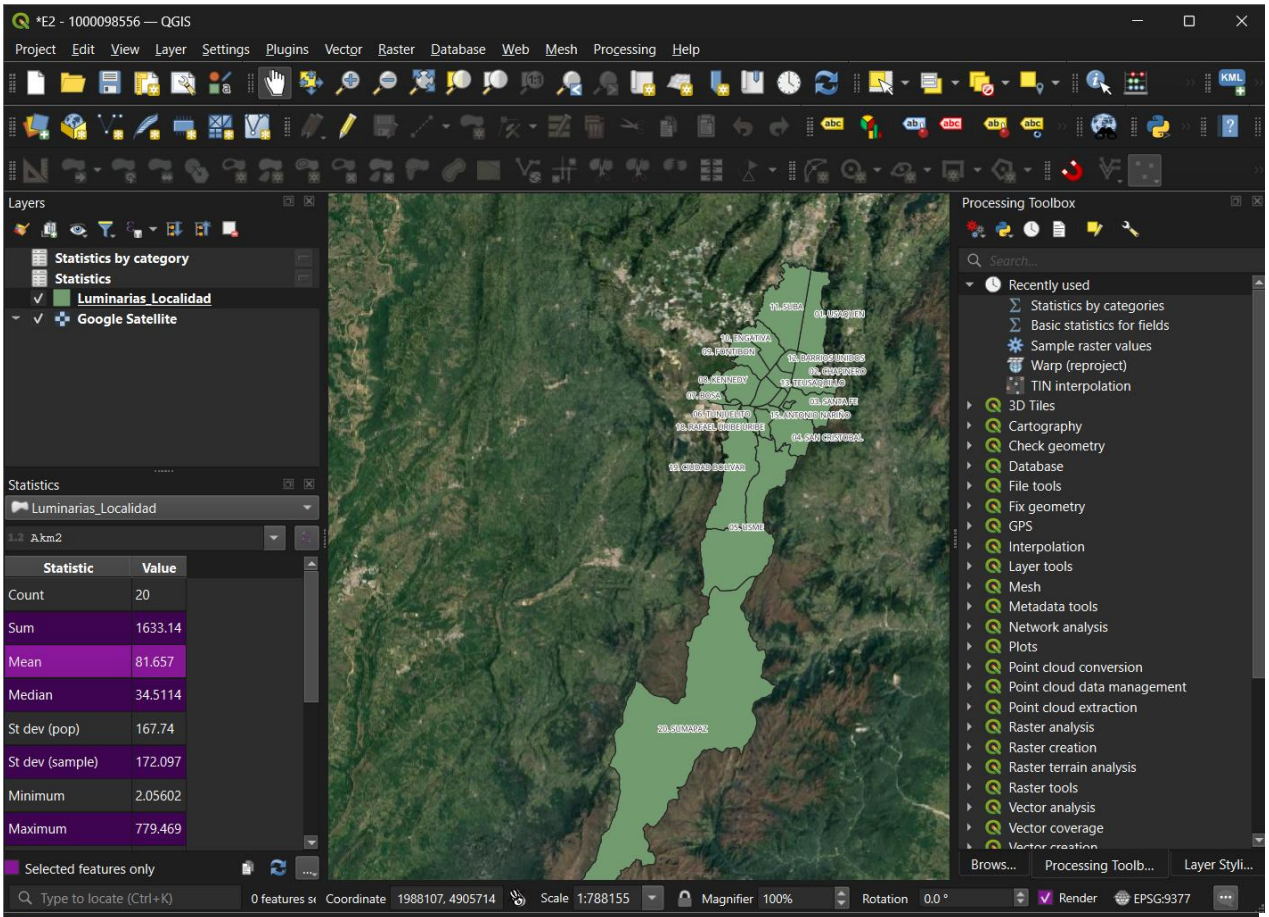


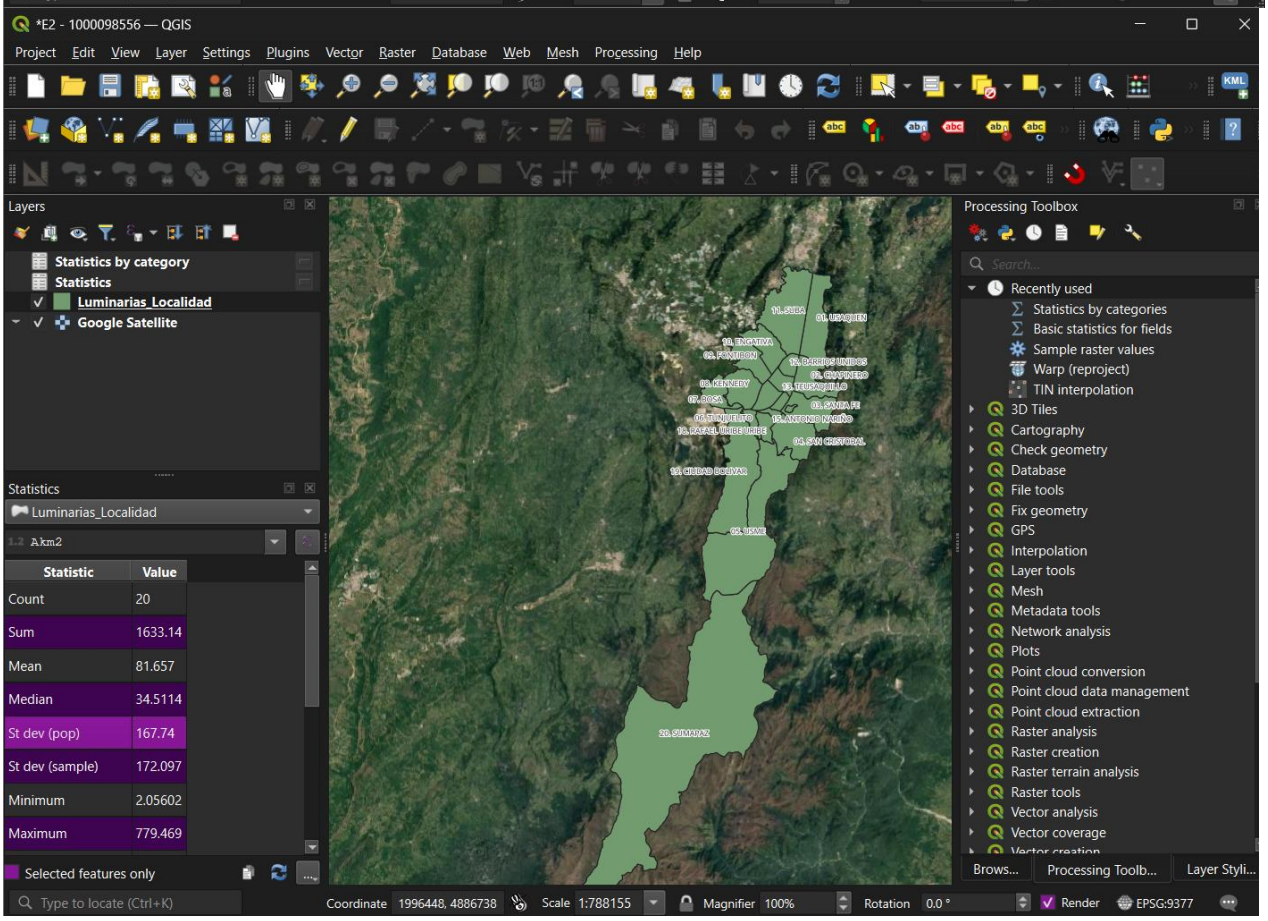
- [x] Sumatoria de áreas por localidad (km²): 1633.1403988712998
- [x] Área (km²) localidad SUMAPAZ: 779.4695
- [x] ADP (%) localidad USME: 13.1426
- [x] ADP (%) localidad SANTA FE: 2.76
- [x] Promedio (mean) de área por localidad (km²): 81.657
- [x] Desviación estándar (St dev pop) de área por localidad (km²): 167.74
- [ ] (No resuelto)

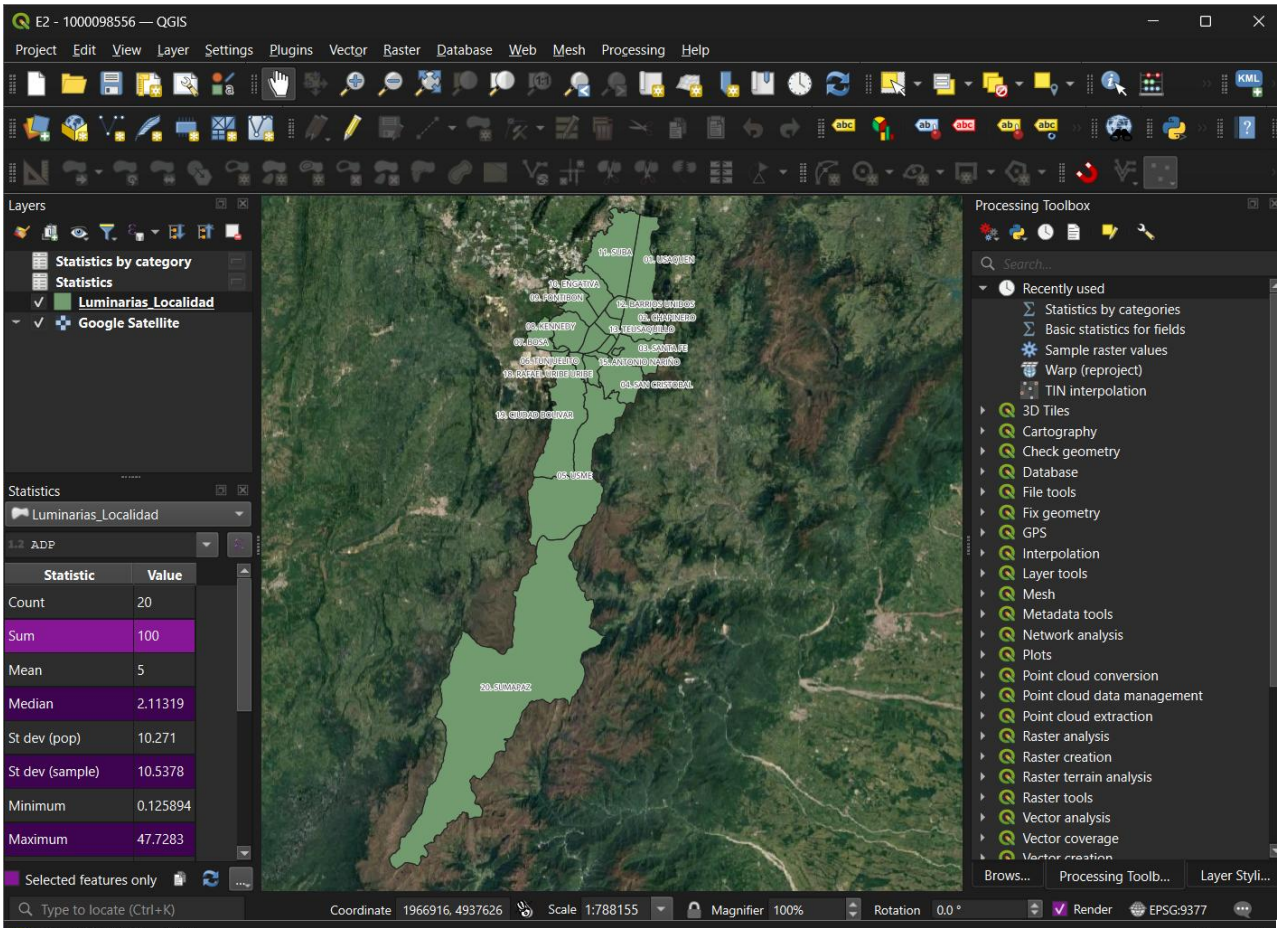
Capturas de pantalla:

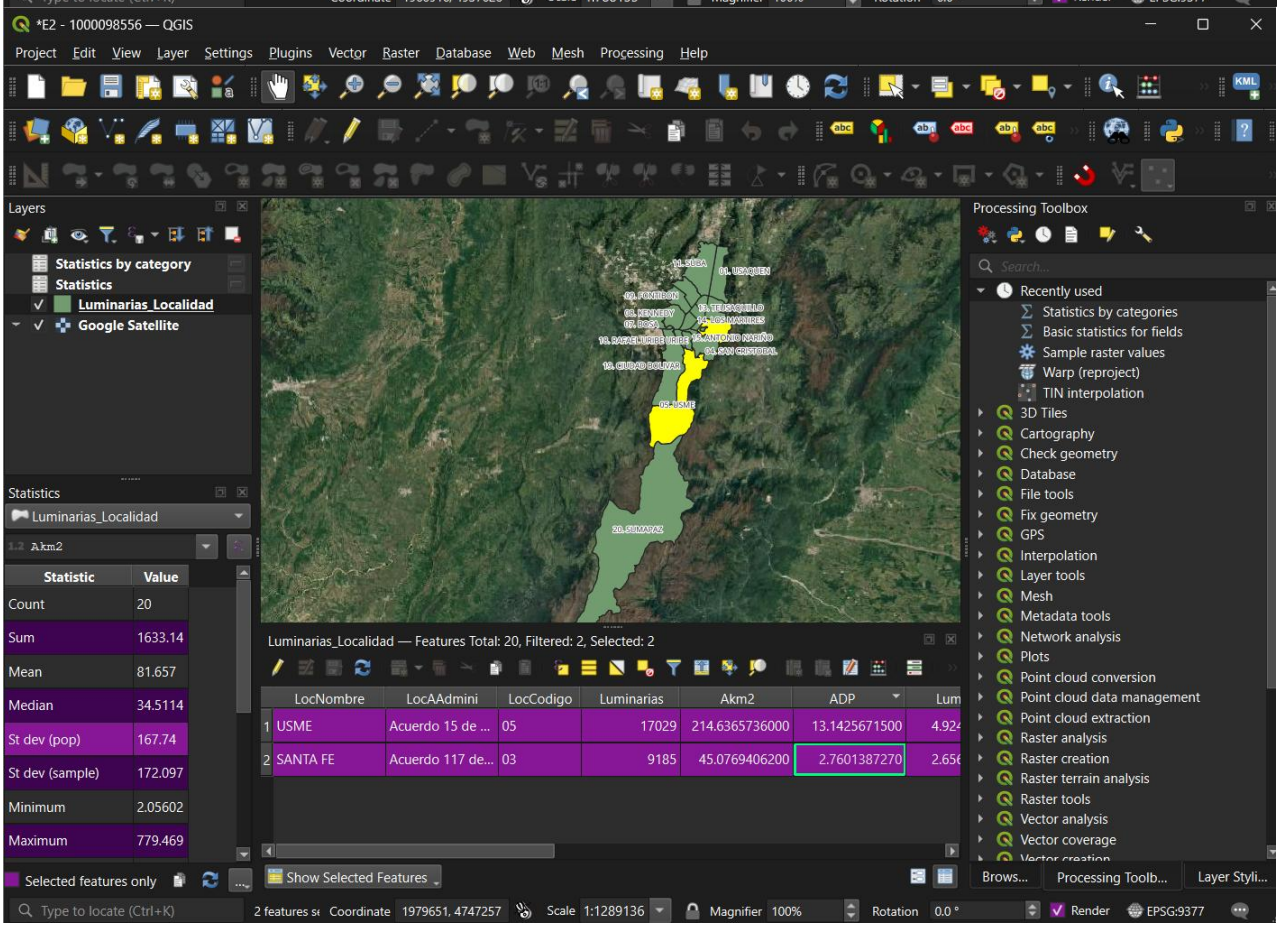

*E2 - 1000098556 — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Layers
Statistics by category
Statistics
Luminarias_Localidad
Google Satellite
Processing Toolbox
basics
Recently used
Vector Analysis - Basic Statistics for Fields
Parameters
Log
Chacon\AppData\Local\Temp\processing_KDbdGc\79b42950af4c44d0acb3bbc32c83c579\OUTPUT_HTML_FILE.html
COUNT: 20
UNIQUE: 20
EMPTY: 0
FILLED: 20
MIN: 2.056020523
MAX: 779.4694708
CV: 2.0541975526804945
SUM: 1633.1403986169996
MEAN: 81.65701993084998
STD_DEV: 167.7396505011344
RANGE: 777.4134502769999
MEDIAN: 34.511414025
MINORITY: 2.056020523
MAJORITY: 2.056020523
FIRSTQUARTILE: 12.842602854999999
THIRDQUARTILE: 57.03170766
IQR: 44.189104805
Complete
Cancel
Advanced
Run as Batch Process...
Change Parameters
Close
Help
Statistics
Luminarias_Localidad
Akm2_
Statistic
Value
Count 20
Sum 1633.14
Mean 81.657
Median 34.5114
St dev (pop) 167.74
St dev (sample) 172.097
Minimum 2.05602
Maximum 779.469
Selected features only
Type to locate (Ctrl+K)
Coordinate 2024496, 4921563
Scale 1:788155
Magnifier 100%
Rotation 0.0 °
Render
EPSG:9377

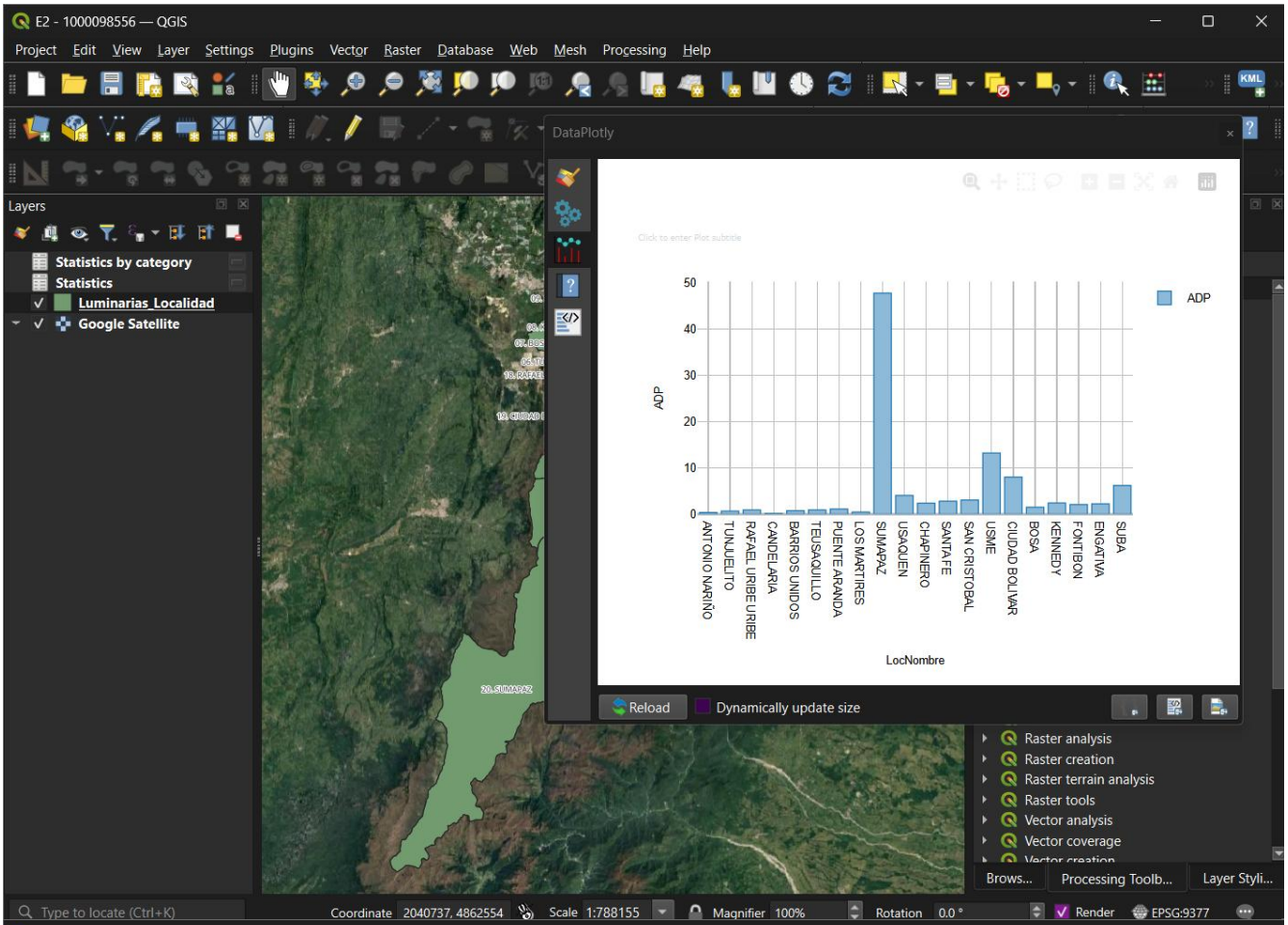

E2 - 1000098556 — QGIS
Project Edit View
Luminarias_Localidad — Features Total: 20, Filtered: 1, Selected: 1
LocNombre
LocAAdmini
LocCodigo
Luminarias
Akm2
ADP
LuminDP
1 SUMAPAZ Acuerdo 9 de 1... 20 NULL 779.4694708000 47.7282584700 NULL

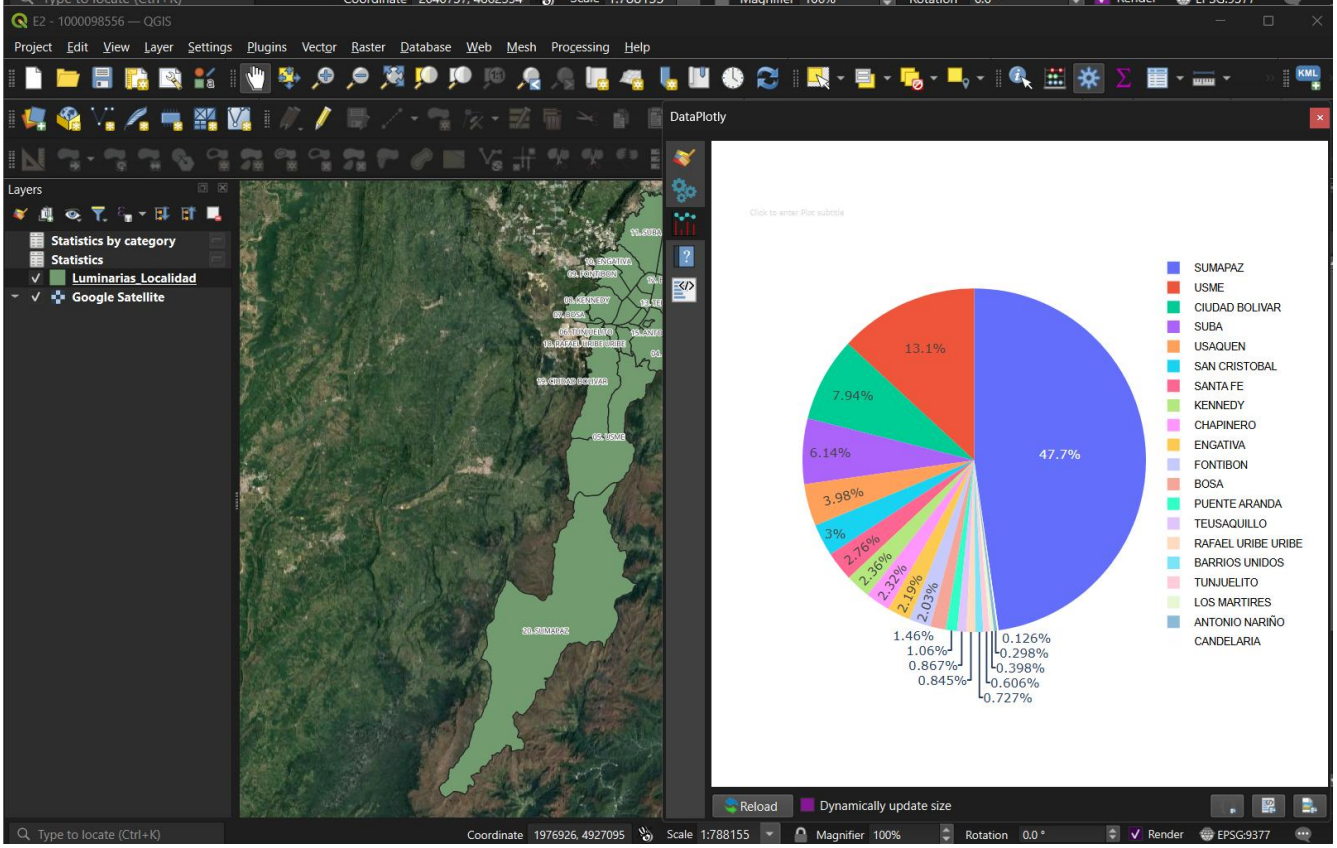

*E2 - 1000098556 — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Layers
Statistics by category
Statistics
Luminarias_Localidad
Google Satellite
Statistics
Luminarias_Localidad
Akm2
Statistic Value
Count 20
Sum 1633.14
Mean 81.657
Median 34.5114
St dev (pop) 167.74
St dev (sample) 172.097
Minimum 2.05602
Maximum 779.469
Selected features only
Type to locate (Ctrl+K)
0 features s
Coordinate 1988107, 4905714
Scale 1:788155
Magnifier 100%
Rotation 0.0 °
Render
EPSG:9377
Processing Toolbox
Search...
Recently used
Statistics by categories
Basic statistics for fields
Sample raster values
Warp (reproject)
TIN interpolation
3D Tiles
Cartography
Check geometry
Database
File tools
Fix geometry
GPS
Interpolation
Layer tools
Mesh
Metadata tools
Network analysis
Plots
Point cloud conversion
Point cloud data management
Point cloud extraction
Raster analysis
Raster creation
Raster terrain analysis
Raster tools
Vector analysis
Vector coverage
Brows...
Processing Toolb...
Layer Styli...


*E2 - 1000098556 — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Layers
Statistics by category
Statistics
Luminarias_Localidad
Google Satellite
Statistics
Luminarias_Localidad
Akm2
Statistic Value
Count 20
Sum 1633.14
Mean 81.657
Median 34.5114
St dev (pop) 167.74
St dev (sample) 172.097
Minimum 2.05602
Maximum 779.469
Selected features only
Type to locate (Ctrl+K)
Coordinate 1996448, 4886738
Scale 1:788155
Magnifier 100%
Rotation 0.0 °
Render
EPSG:9377
Processing Toolbox
Search...
Recently used
Statistics by categories
Basic statistics for fields
Sample raster values
Warp (reproject)
TIN interpolation
3D Tiles
Cartography
Check geometry
Database
File tools
Fix geometry
GPS
Interpolation
Layer tools
Mesh
Metadata tools
Network analysis
Plots
Point cloud conversion
Point cloud data management
Point cloud extraction
Raster analysis
Raster creation
Raster terrain analysis
Raster tools
Vector analysis
Vector coverage
Brows...
Processing Toolb...
Layer Styli...


E2 - 1000098556 — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Layers
Statistics by category
Statistics
Luminarias_Localidad
Google Satellite
Statistics
Luminarias_Localidad
ADP
Statistic Value
Count 20
Sum 100
Mean 5
Median 2.11319
St dev (pop) 10.271
St dev (sample) 10.5378
Minimum 0.125894
Maximum 47.7283
Selected features only
Processing Toolbox
Recently used
Statistics by categories
Basic statistics for fields
Sample raster values
Warp (reproject)
TIN interpolation
3D Tiles
Cartography
Check geometry
Database
File tools
Fix geometry
GPS
Interpolation
Layer tools
Mesh
Metadata tools
Network analysis
Plots
Point cloud conversion
Point cloud data management
Point cloud extraction
Raster analysis
Raster creation
Raster terrain analysis
Raster tools
Vector analysis
Vector coverage
Brows...
Processing Toolb...
Layer Styli...
Type to locate (Ctrl+K)
Coordinate 1966916, 4937626
Scale 1:788155
Magnifier 100%
Rotation 0.0 °
Render
EPSG:9377


*E2 - 1000098556 — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Layers
Statistics by category
Statistics
Luminarias_Localidad
Google Satellite
Statistics
Luminarias_Localidad
Akm2
Statistic Value
Count 20
Sum 1633.14
Mean 81.657
Median 34.5114
St dev (pop) 167.74
St dev (sample) 172.097
Minimum 2.05602
Maximum 779.469
Selected features only
Luminarias_Localidad — Features Total: 20, Filtered: 2, Selected: 2
LocNombre LocAAdmini LocCodigo Luminarias Akm2 ADP
1 USME Acuerdo 15 de ... 05 17029 214.6365736000 13.1425671500
2 SANTA FE Acuerdo 117 de... 03 9185 45.0769406200 2.7601387270
Show Selected Features
Processing Toolbox
Recently used
Statistics by categories
Basic statistics for fields
Sample raster values
Warp (reproject)
TIN interpolation
3D Tiles
Cartography
Check geometry
Database
File tools
Fix geometry
GPS
Interpolation
Layer tools
Mesh
Metadata tools
Network analysis
Plots
Point cloud conversion
Point cloud data management
Point cloud extraction
Raster analysis
Raster creation
Raster terrain analysis
Raster tools
Vector analysis
Vector coverage
Brows...
Processing Toolb...
Layer Styli...
Type to locate (Ctrl+K)
2 features se
Coordinate 1979651, 4747257
Scale 1:1289136
Magnifier 100%
Rotation 0.0 °
Render
EPSG:9377


E2 - 1000098556 — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
DataPlotly
Layers
Statistics by category
Statistics
Luminarias_Localidad
Google Satellite
ADP
50
40
30
20
10
0
ANTONIO NARIÑO
TUNJUELITO
RAFAEL URIBE URIBE
CANDELARIA
BARRIOS UNIDOS
TEUSAQUILLO
PUENTE ARANDA
LOS MARTIRES
SUMAPAZ
USAQUEN
CHAPINERO
SANTA FE
SAN CRISTOBAL
USME
CIUDAD BOLIVAR
BOSA
KENNEDY
FONTIBON
ENGATIVA
SUBA
LocNombre
Reload
Dynamically update size
Raster analysis
Raster creation
Raster terrain analysis
Raster tools
Vector analysis
Vector coverage
Brows...
Processing Toolb...
Layer Styli...
Type to locate (Ctrl+K)
Coordinate 2040737, 4862554
Scale 1:788155
Magnifier 100%
Rotation 0.0 °
Render
EPSG:9377


E2 - 1000098556 — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
DataPlotly
Layers
Statistics by category
Statistics
Luminarias_Localidad
Google Satellite
SUMAPAZ
USME
CIUDAD BOLIVAR
SUBA
USAQUEN
SAN CRISTOBAL
SANTA FE
KENNEDY
CHAPINERO
ENGATIVA
FONTIBON
BOSA
PUENTE ARANDA
TEUSAQUILLO
RAFAEL URIBE URIBE
BARRIOS UNIDOS
TUNJUELITO
LOS MARTIRES
ANTONIO NARIÑO
CANDELARIA
47.7%
13.1%
7.94%
6.14%
3.98%
3%
2.76%
2.36%
2.32%
2.19%
2.03%
1.46%
1.06%
0.867%
0.845%
0.606%
0.727%
0.398%
0.298%
0.126%
Reload
Dynamically update size
Type to locate (Ctrl+K)
Coordinate 1976926, 4927095
Scale 1:788155
Magnifier 100%
Rotation 0.0 °
Render
EPSG:9377

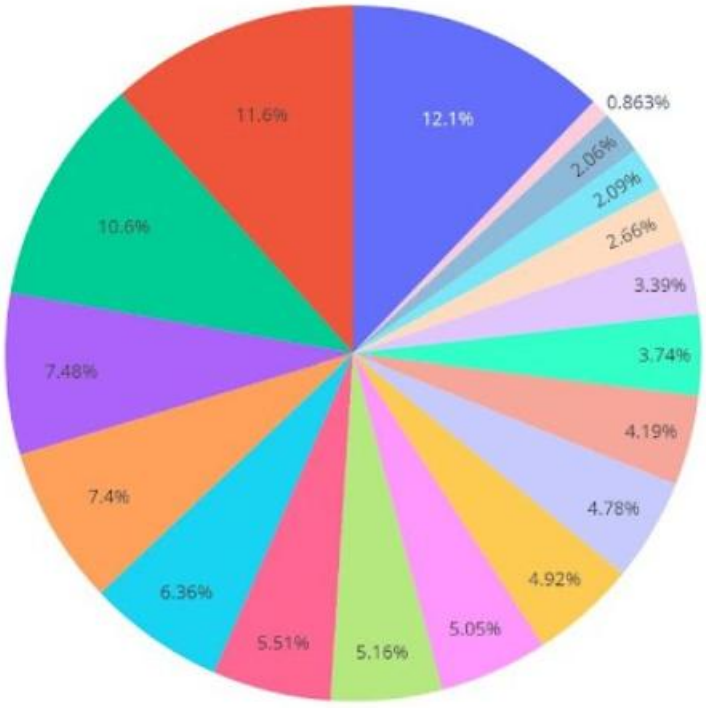
3. P3

Texto de la actividad:

3. Distribución porcentual de Luminarias: cree el campo *LuminDP* real o double (10 posiciones enteras y 10 decimales de precisión) y calcule la distribución porcentual de luminarias por localidad. * 4 puntos

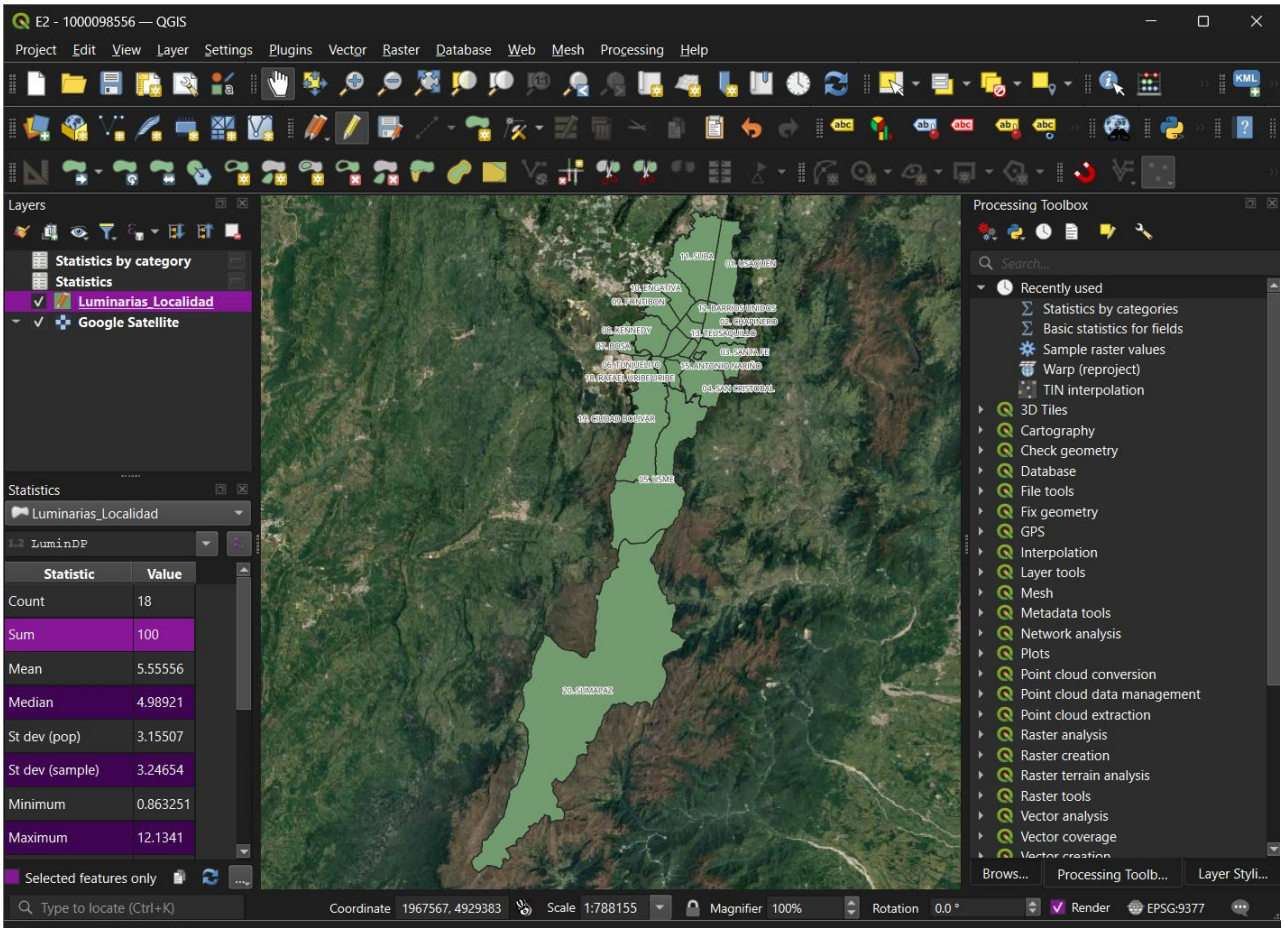
Procedimiento:

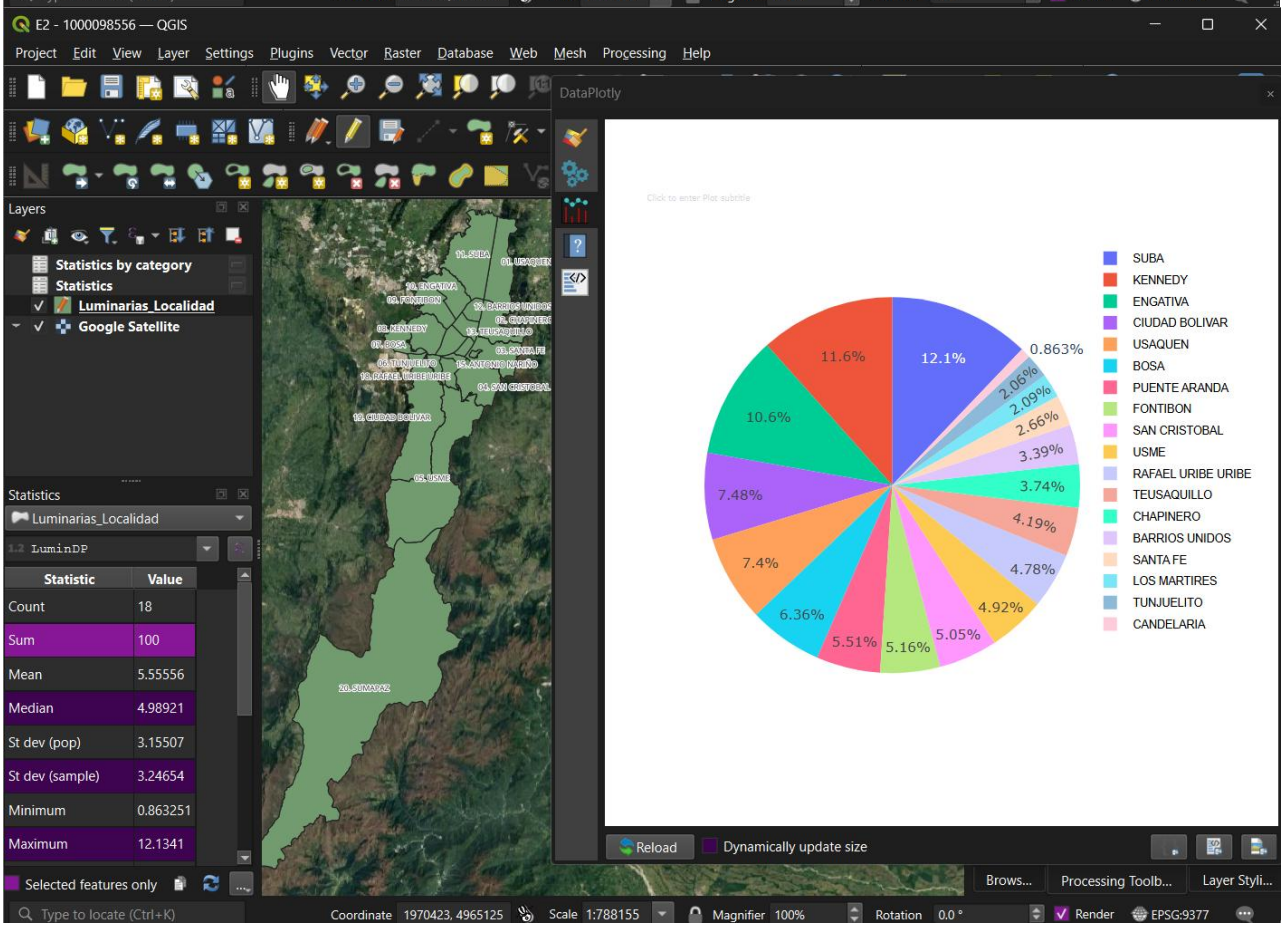
- Para el cálculo del campo *LuminDP*, divida el número de luminarias de cada localidad entre el total de luminarias de Bogotá D.C.
- Verifique que la sumatoria de las distribuciones porcentuales sea 100%
- Utilizando el complemento *DataPlotly* o la herramienta *Processing Toolbox / Plots / Bar Plot*, cree una gráfica de pastel o de barras representando las distribuciones porcentuales obtenidas por localidad
- Esta pregunta solo será válida si realiza los cálculos solicitados en QGIS

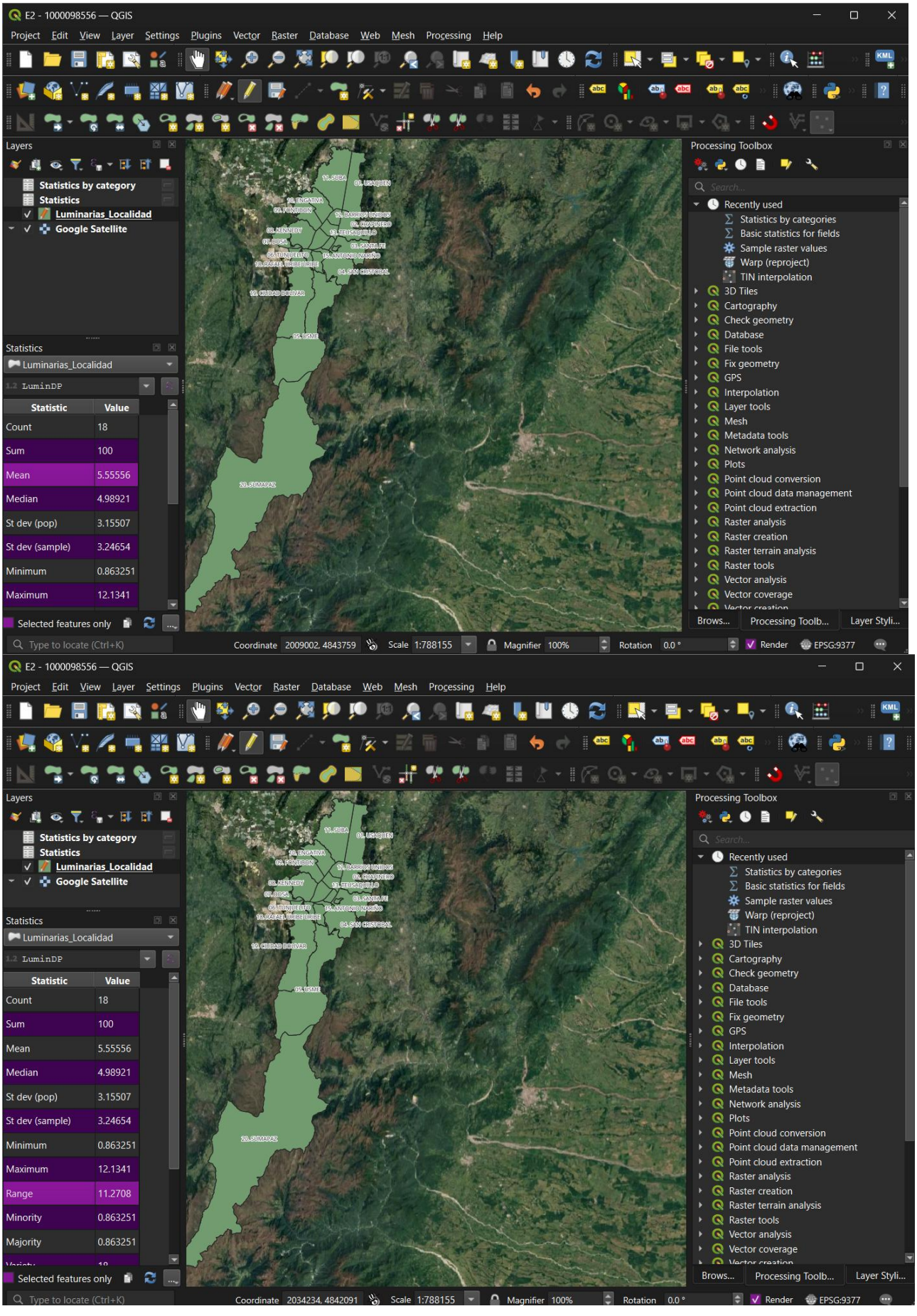


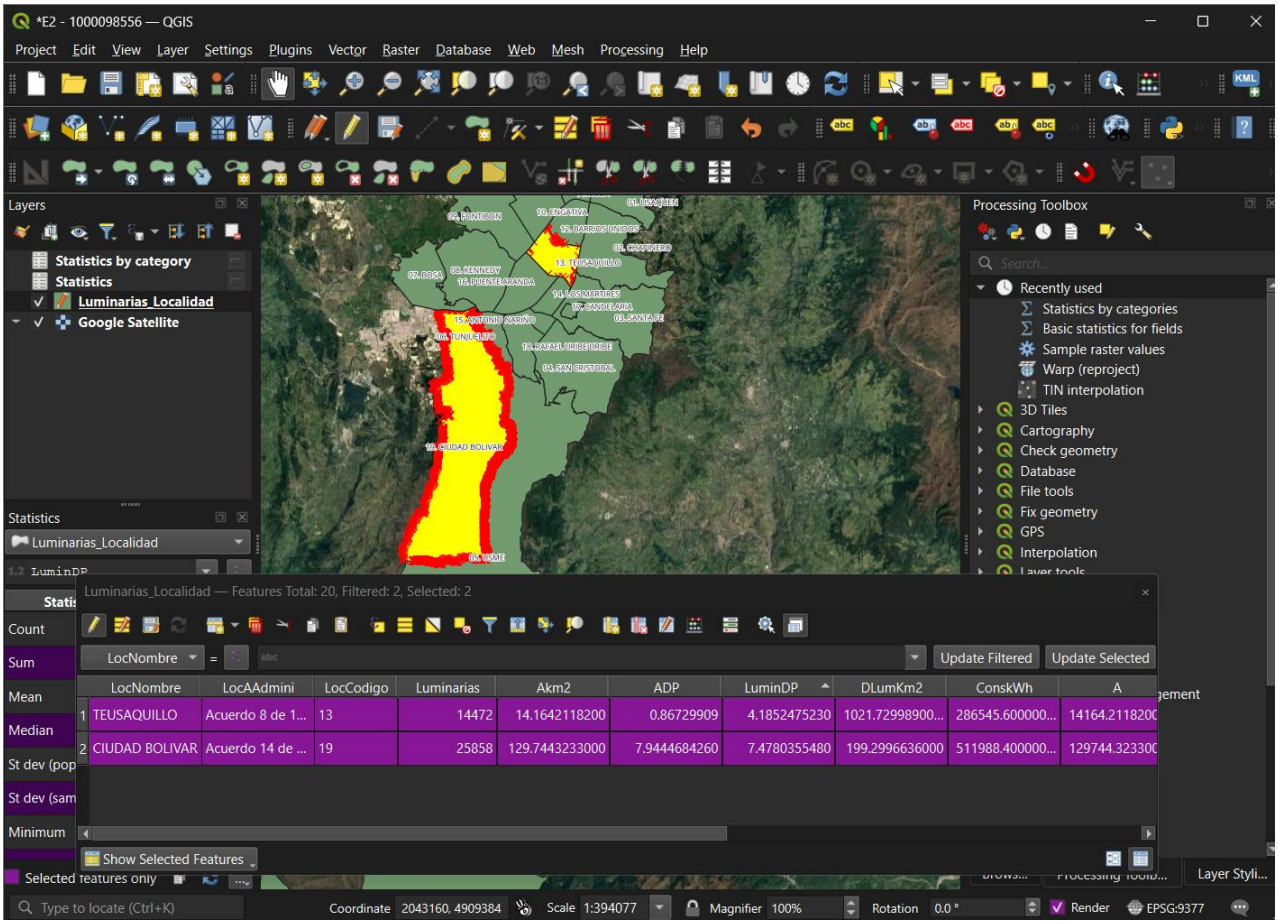
- [x] % luminarias CIUDAD BOLIVAR: 7.4780355480
- [x] % luminarias TEUSAQUILLO: 4.1852475230
- [x] % promedio (mean) de luminarias por localidad: 5.55556
- [x] Rango (range) de % de luminarias por localidad: 11.2708
- [ ] (No resuelto)

Capturas de pantalla:


E2 - 1000098556 — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Layers
Statistics by category
Statistics
Luminarias_Localidad
Google Satellite
Processing Toolbox
Recently used
Statistics by categories
Basic statistics for fields
Sample raster values
Warp (reproject)
TIN interpolation
3D Tiles
Cartography
Check geometry
Database
File tools
Fix geometry
GPS
Interpolation
Layer tools
Mesh
Metadata tools
Network analysis
Plots
Point cloud conversion
Point cloud data management
Point cloud extraction
Raster analysis
Raster creation
Raster terrain analysis
Raster tools
Vector analysis
Vector coverage
Statistics
Luminarias_Localidad
LuminDP
Statistic Value
Count 18
Sum 100
Mean 5.55556
Median 4.98921
St dev (pop) 3.15507
St dev (sample) 3.24654
Minimum 0.863251
Maximum 12.1341
Selected features only
Type to locate (Ctrl+K)
Coordinate 1967567, 4929383
Scale 1:788155
Magnifier 100%
Rotation 0.0 °
Render
EPSG:9377


E2 - 1000098556 — QGIS
DataPlotly
SUBA
KENNEDY
ENGATIVA
CIUDAD BOLIVAR
USAQUEN
BOSA
PUENTE ARANDA
FONTIBON
SAN CRISTOBAL
USME
RAFAEL URIBE URIBE
TEUSAQUILLO
CHAPINERO
BARRIOS UNIDOS
SANTA FE
LOS MARTIRES
TUNJUELITO
CANDELARIA
12.1%
11.6%
10.6%
7.48%
7.4%
6.36%
5.51%
5.16%
5.05%
4.92%
4.78%
4.19%
3.74%
3.39%
2.66%
2.09%
2.06%
0.863%
Reload
Dynamically update size
Coordinate 1970423, 4965125
Scale 1:788155
Magnifier 100%
Rotation 0.0 °
Render
EPSG:9377


E2 - 1000098556 — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Layers
Statistics by category
Statistics
Luminarias_Localidad
Google Satellite
Statistics
Luminarias_Localidad
LuminDP
Statistic Value
Count 18
Sum 100
Mean 5.55556
Median 4.98921
St dev (pop) 3.15507
St dev (sample) 3.24654
Minimum 0.863251
Maximum 12.1341
Selected features only
Processing Toolbox
Recently used
Statistics by categories
Basic statistics for fields
Sample raster values
Warp (reproject)
TIN interpolation
3D Tiles
Cartography
Check geometry
Database
File tools
Fix geometry
GPS
Interpolation
Layer tools
Mesh
Metadata tools
Network analysis
Plots
Point cloud conversion
Point cloud data management
Point cloud extraction
Raster analysis
Raster creation
Raster terrain analysis
Raster tools
Vector analysis
Vector coverage
Brows...
Processing Toolb...
Layer Styli...
Type to locate (Ctrl+K)
Coordinate 2009002, 4843759
Scale 1:788155
Magnifier 100%
Rotation 0.0 °
Render
EPSG:9377
E2 - 1000098556 — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Layers
Statistics by category
Statistics
Luminarias_Localidad
Google Satellite
Statistics
Luminarias_Localidad
LuminDP
Statistic Value
Count 18
Sum 100
Mean 5.55556
Median 4.98921
St dev (pop) 3.15507
St dev (sample) 3.24654
Minimum 0.863251
Maximum 12.1341
Range 11.2708
Minority 0.863251
Majority 0.863251
Selected features only
Processing Toolbox
Recently used
Statistics by categories
Basic statistics for fields
Sample raster values
Warp (reproject)
TIN interpolation
3D Tiles
Cartography
Check geometry
Database
File tools
Fix geometry
GPS
Interpolation
Layer tools
Mesh
Metadata tools
Network analysis
Plots
Point cloud conversion
Point cloud data management
Point cloud extraction
Raster analysis
Raster creation
Raster terrain analysis
Raster tools
Vector analysis
Vector coverage
Brows...
Processing Toolb...
Layer Styli...
Type to locate (Ctrl+K)
Coordinate 2034234, 4842091
Scale 1:788155
Magnifier 100%
Rotation 0.0 °
Render
EPSG:9377



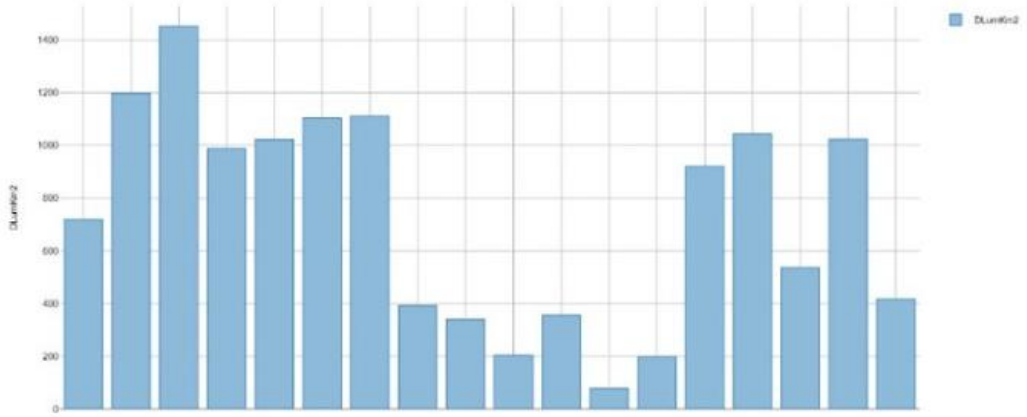
4. P4

Texto de la actividad:

4. Densidad de Luminarias por localidad: cree el campo *DLumKm2* real o double (10 posiciones enteras y 10 decimales de precisión) y calcule la densidad de luminarias / km² de cada localidad * 4 puntos

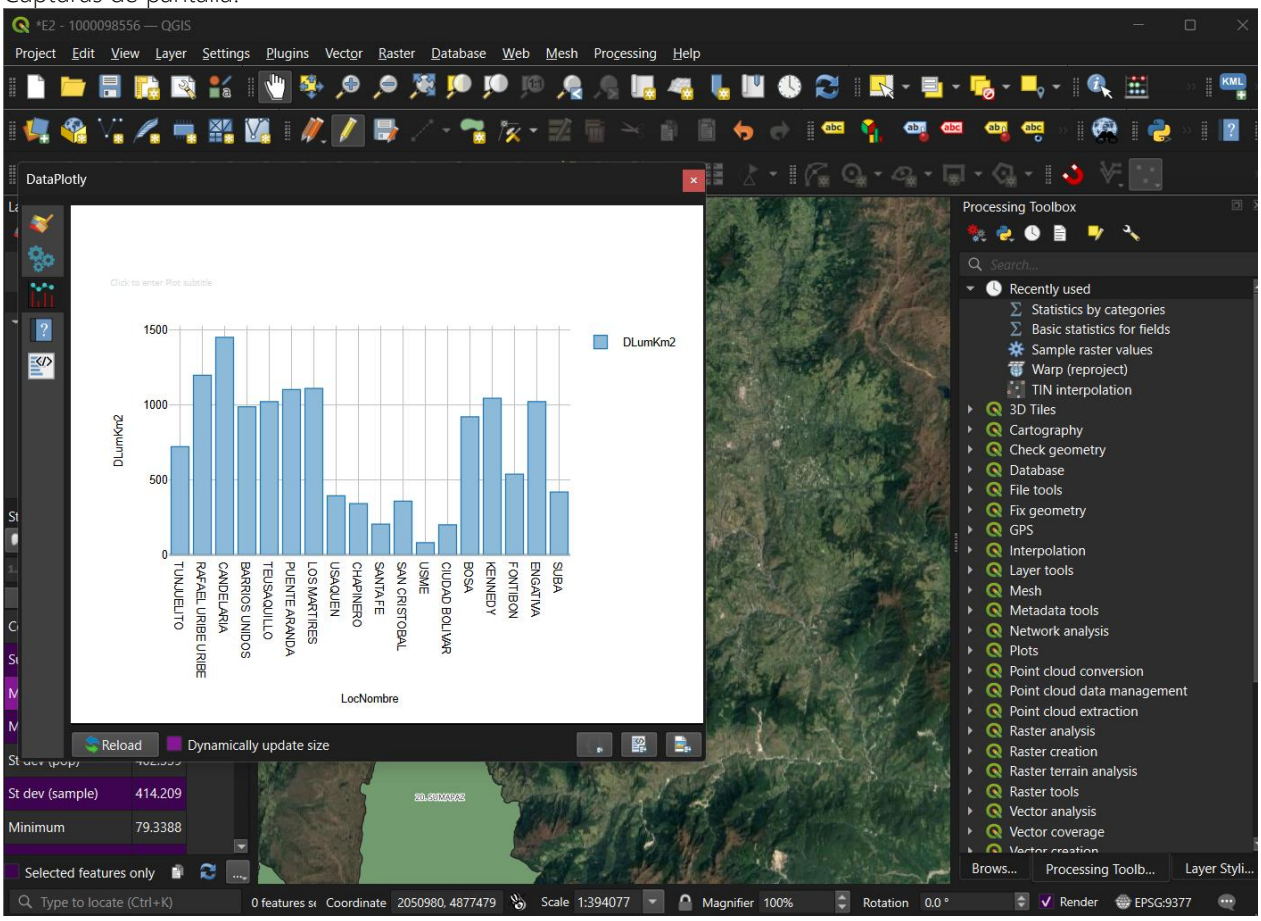
Procedimiento:

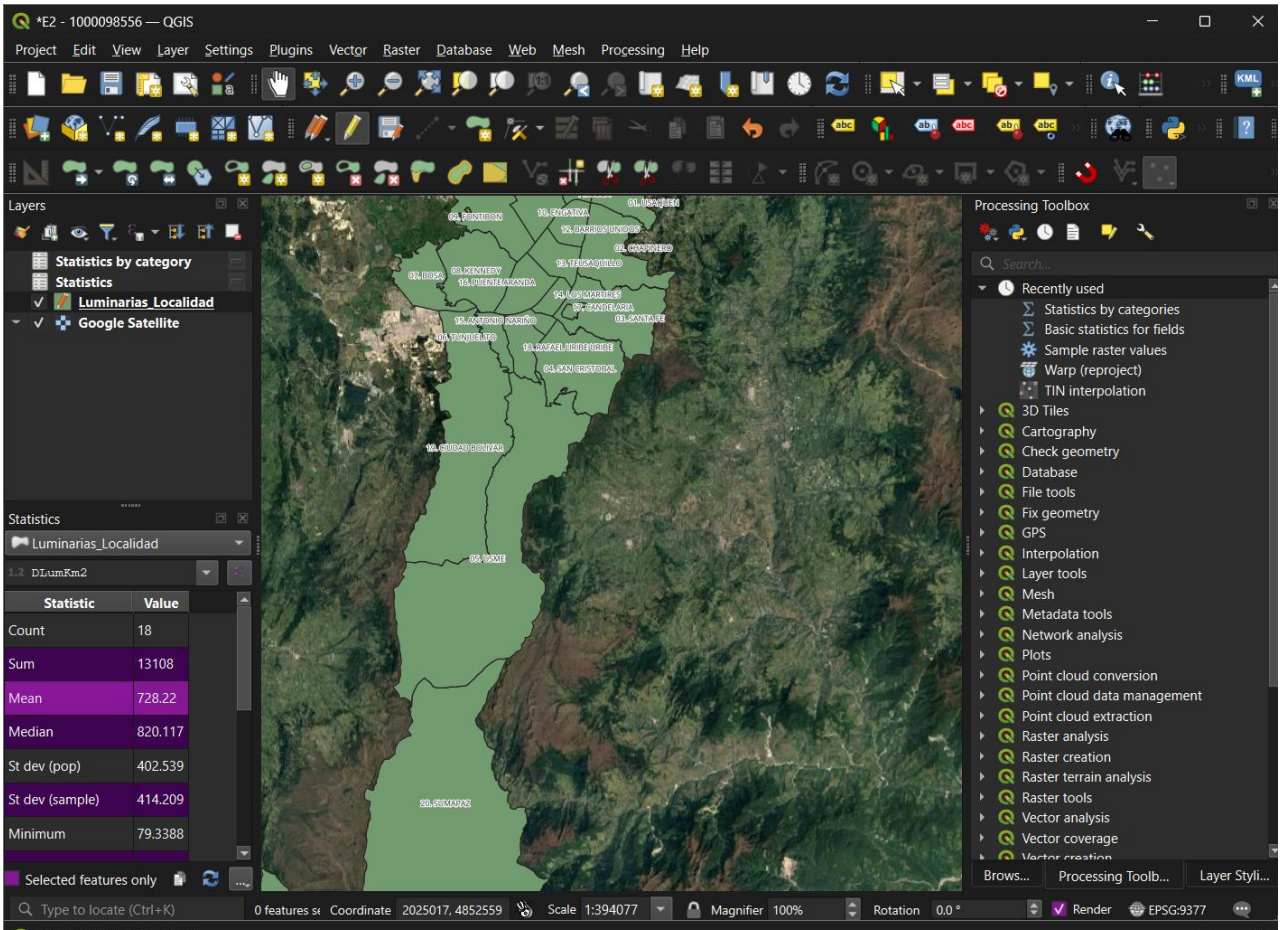
- Para el cálculo del campo *DLumKm2*, divida el número de luminarias de cada localidad entre su propia área en km²
- Utilizando el complemento *DataPlotly* o la herramienta *Processing Toolbox / Plots / Bar Plot*, cree una gráfica de barras representando las densidades obtenidas por localidad
- Esta pregunta solo será válida si realiza los cálculos solicitados en QGIS

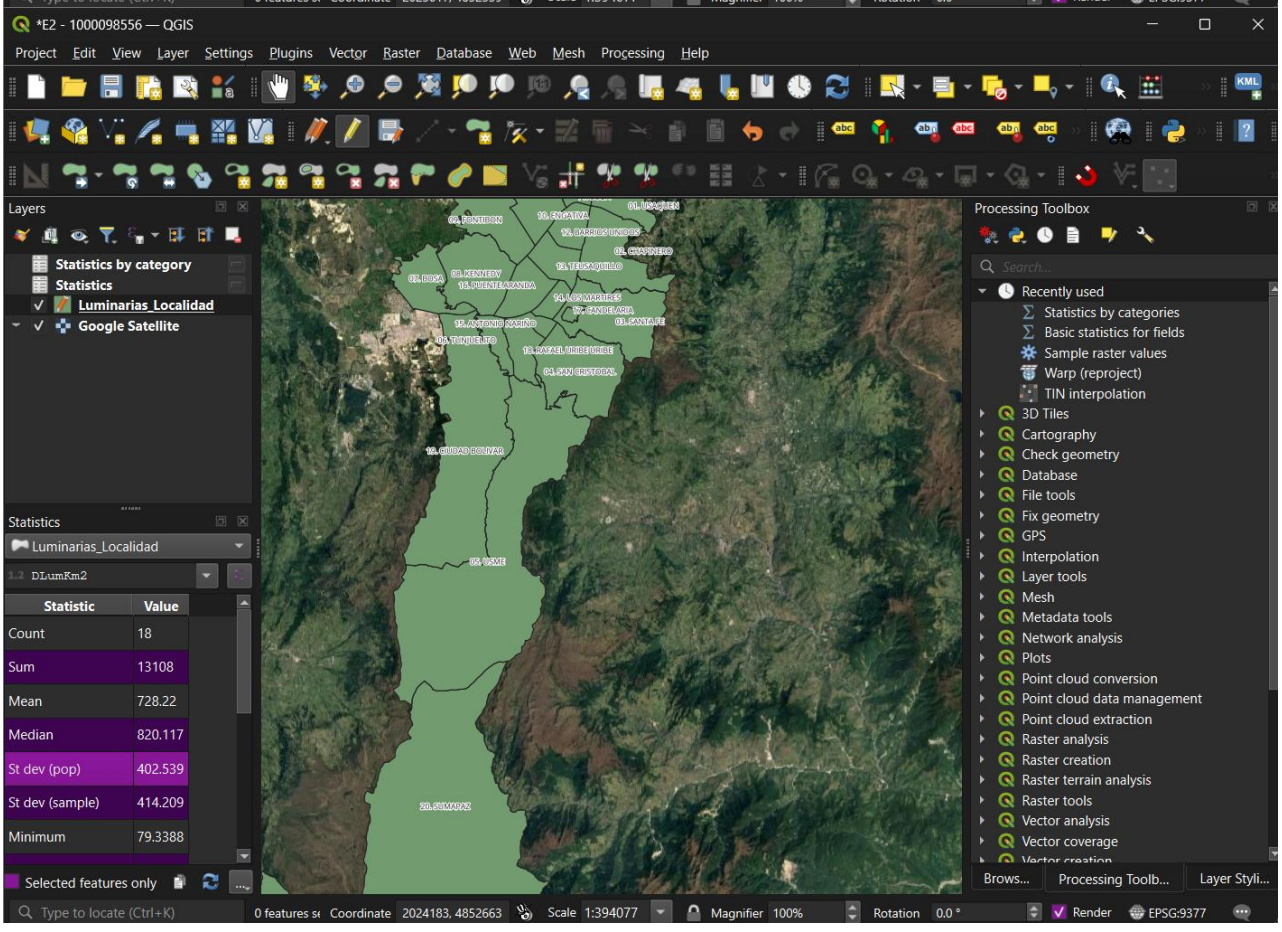


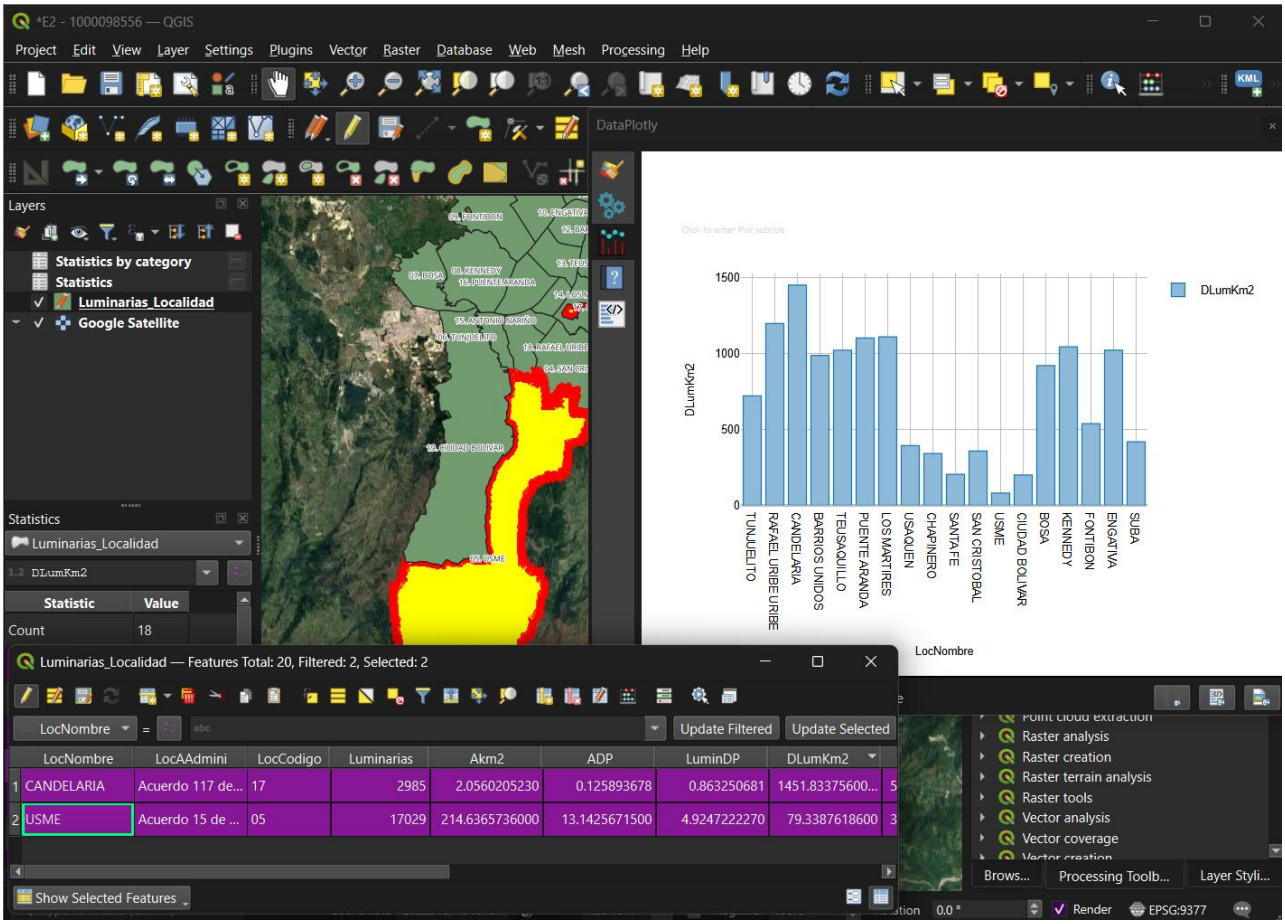
- [x] Localidad con mayor densidad (lum/km²): CANDELARIA con 1451.8337568170
- [x] Localidad con menor densidad (lum/km²): USME con 79.3387618633
- [x] Promedio (mean) de densidad (lum/km²): 728.22
- [x] Desviación estándar (St dev pop) de densidad (lum/km²): 402.539
- [ ] (No resuelto)

Capturas de pantalla:




*E2 - 1000098556 — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Layers
Statistics by category
Statistics
Luminarias_Localidad
Google Satellite
Statistics
Luminarias_Localidad
DLumKm2
Statistic Value
Count 18
Sum 13108
Mean 728.22
Median 820.117
St dev (pop) 402.539
St dev (sample) 414.209
Minimum 79.3388
Selected features only
Processing Toolbox
Recently used
Statistics by categories
Basic statistics for fields
Sample raster values
Warp (reproject)
TIN interpolation
3D Tiles
Cartography
Check geometry
Database
File tools
Fix geometry
GPS
Interpolation
Layer tools
Mesh
Metadata tools
Network analysis
Plots
Point cloud conversion
Point cloud data management
Point cloud extraction
Raster analysis
Raster creation
Raster terrain analysis
Raster tools
Vector analysis
Vector coverage
Brows...
Processing Toolb...
Layer Styli...
Type to locate (Ctrl+K)
0 features se
Coordinate 2025017, 4852559
Scale 1:394077
Magnifier 100%
Rotation 0.0 °
Render
EPSG:9377


*E2 - 1000098556 — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Layers
Statistics by category
Statistics
Luminarias_Localidad
Google Satellite
Statistics
Luminarias_Localidad
DLumKm2
Statistic Value
Count 18
Sum 13108
Mean 728.22
Median 820.117
St dev (pop) 402.539
St dev (sample) 414.209
Minimum 79.3388
Selected features only
Processing Toolbox
Recently used
Statistics by categories
Basic statistics for fields
Sample raster values
Warp (reproject)
TIN interpolation
3D Tiles
Cartography
Check geometry
Database
File tools
Fix geometry
GPS
Interpolation
Layer tools
Mesh
Metadata tools
Network analysis
Plots
Point cloud conversion
Point cloud data management
Point cloud extraction
Raster analysis
Raster creation
Raster terrain analysis
Raster tools
Vector analysis
Vector coverage
Brows...
Processing Toolb...
Layer Styli...
Type to locate (Ctrl+K)
0 features se
Coordinate 2024183, 4852663
Scale 1:394077
Magnifier 100%
Rotation 0.0 °
Render
EPSG:9377



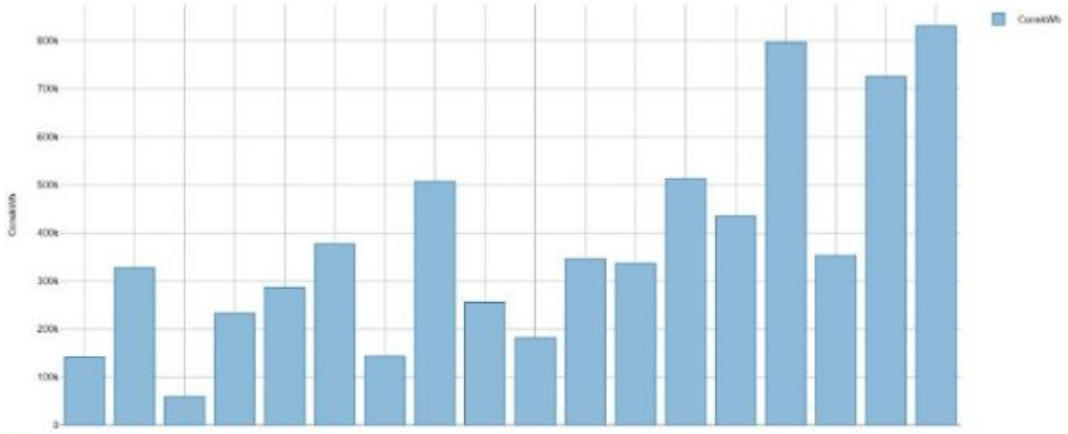
5. P5

Texto de la actividad:

5. Consumo eléctrico mensual: cree el campo *ConskWh* real o double (10 posiciones enteras y 10 decimales de precisión) y calcule el consumo mensual para luminarias que consumen 60 watts o vatios hora durante 11 horas diarias y 30 días en promedio por mes * 3 puntos

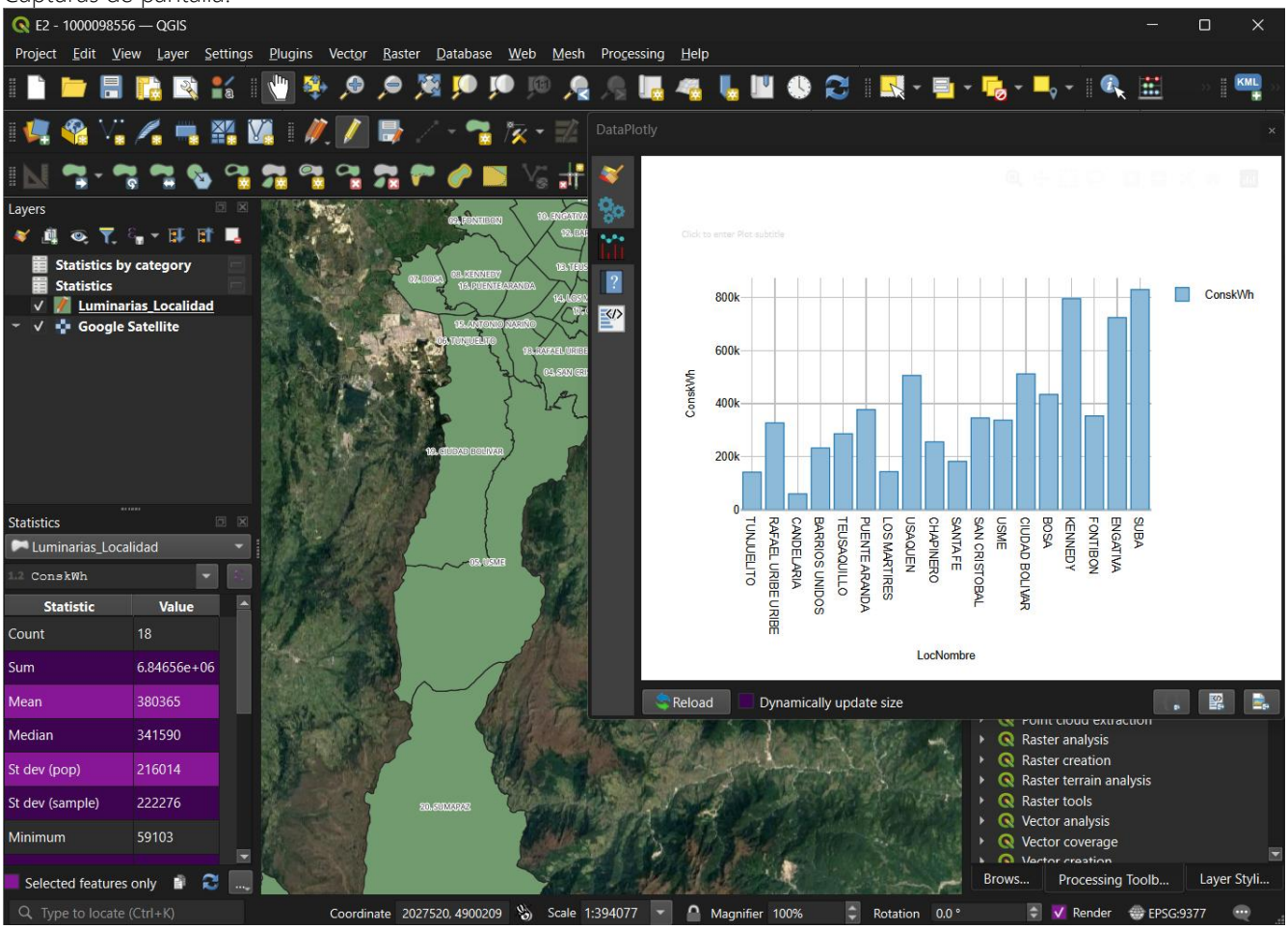
Procedimiento:

- Para el cálculo del campo *ConskWh*, multiplique el número de luminarias * el consumo * el número de horas cada día * el número de días en el mes y divida el resultado en 1000 para obtener los valores en kWh
  . *ConskWh*= "Luminarias" * 60 * 11 * 30 / 1000
- Utilizando el complemento *DataPlotly* o la herramienta *Processing Toolbox / Plots / Bar Plot*, cree una gráfica de barras representando los consumos eléctricos obtenidos por localidad
- Esta pregunta solo será válida si realiza los cálculos solicitados en QGIS



- [x] ConskWh de CHAPINERO: 255776.4
- [x] Promedio (mean) de ConskWh : 380365
- [x] Desviación estándar (St dev pop) de densidad (lum/km²): 216014
- [ ] (No resuelto)

Capturas de pantalla:



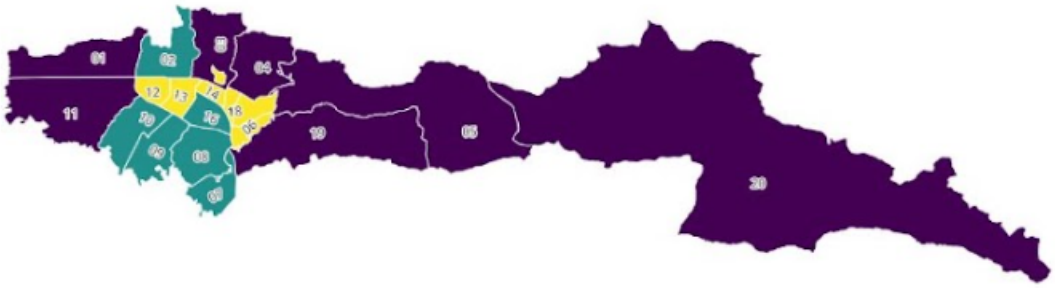
6. P6

Texto de la actividad:

6. Potencial fotovoltáico por localidad: utilizando el 0.001% del área planar de cada localidad, estime el número de páneles solares que pueden ser instalados y el total de energía potencial y real que puede ser producida * 4 puntos

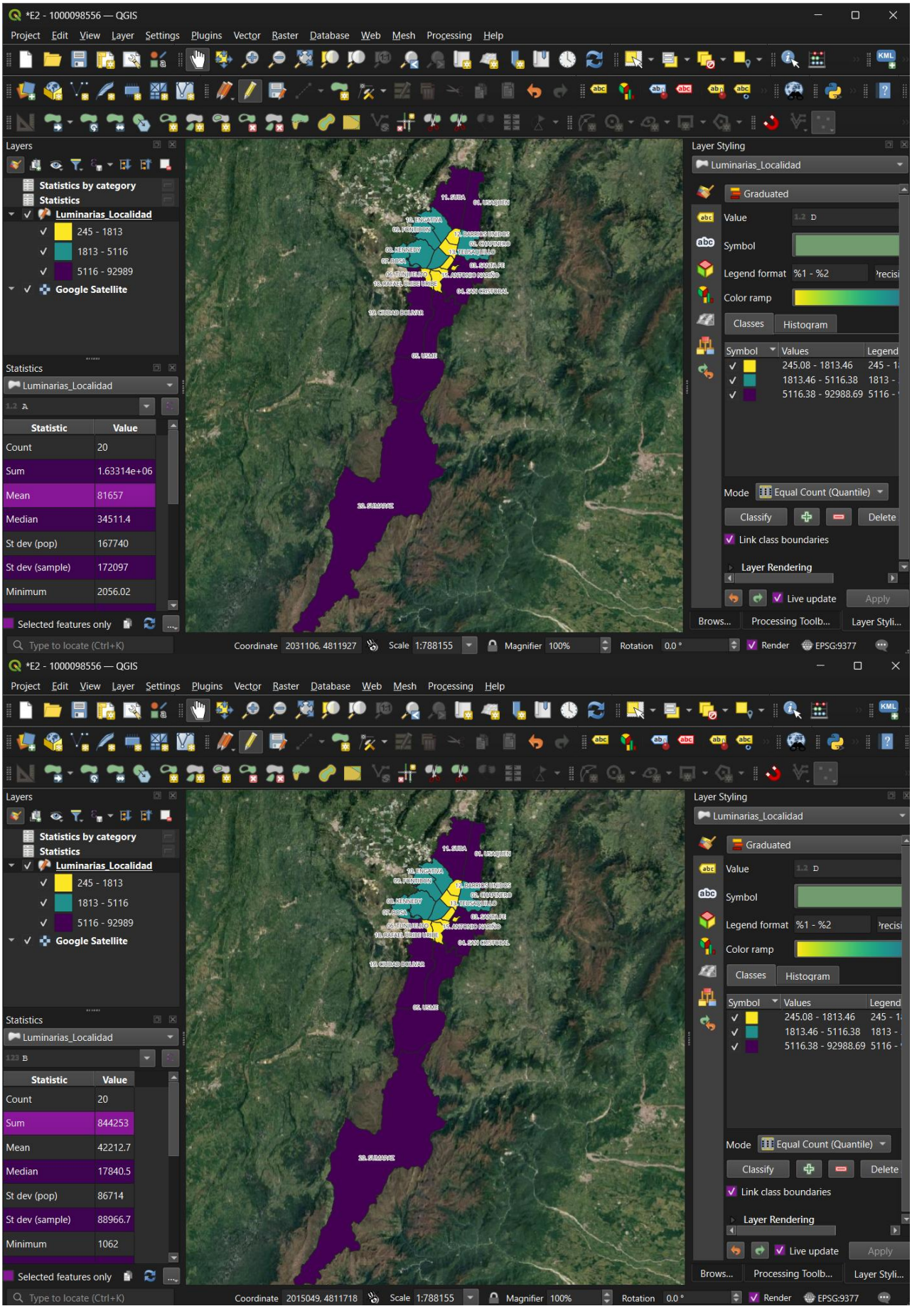
Procedimiento:

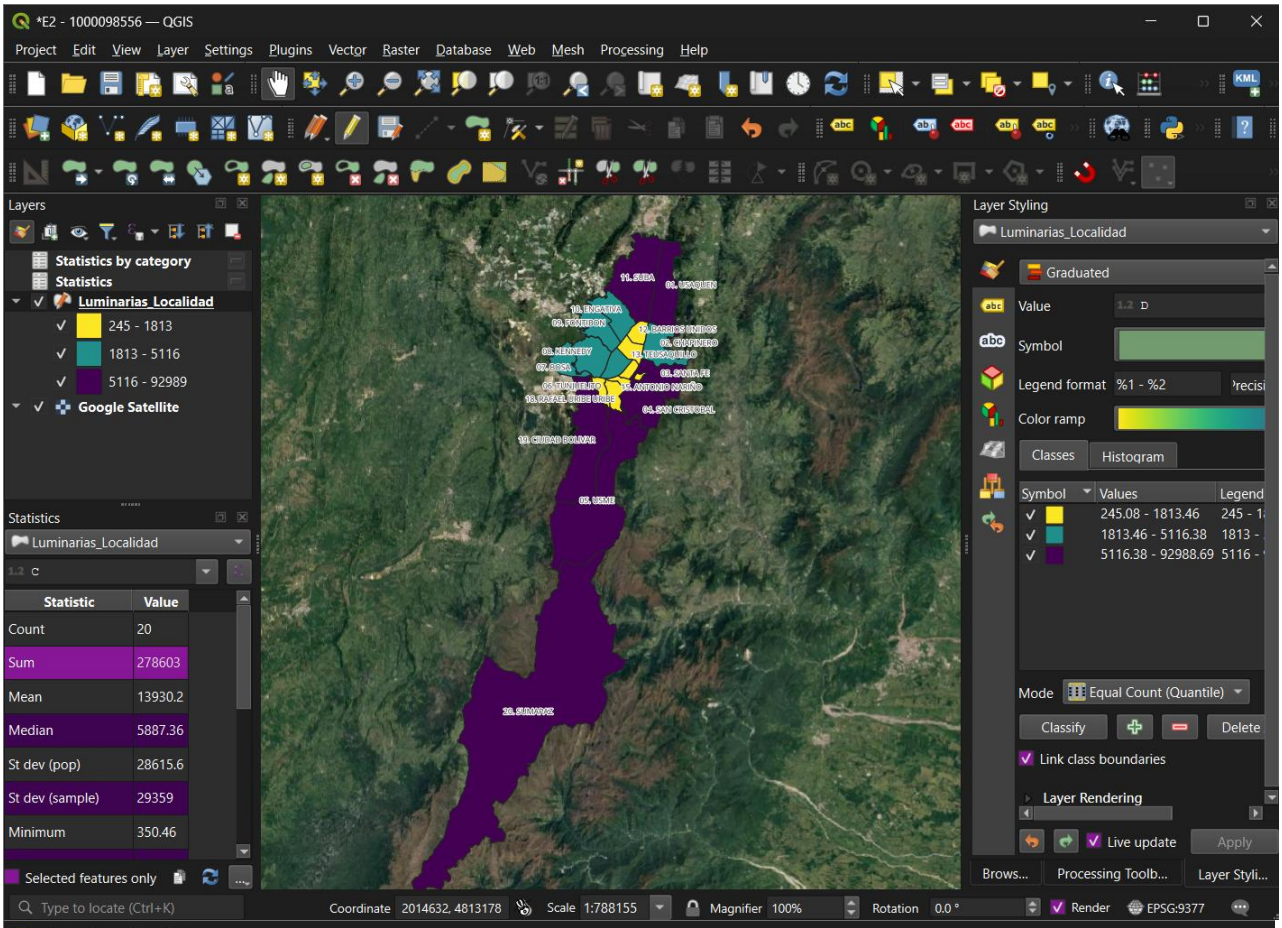
- Cree el campo tipo real *A* (10 posiciones enteras y 10 decimales de precisión) y calcule el área en m² usable por localidad. *A = area(@geometry) * 0.001*
- *Cree el campo tipo entero B* y calcule el número de páneles solares de 1950 * 992 mm que pueden ser instalados: *B = floor("A"/((1950*992)/(1000*1000)))*
- *Cree el campo tipo real* C (10 posiciones enteras y 10 decimales de precisión) y calcule el p*otencial de Kilovatios producidos por hora (Kwh): C = "B" * 330 / 1000*. Potencia por panel: 330 watts-hora
- *Cree el campo tipo real D (10 posiciones enteras y 10 decimales de precisión)* y calcule el real de Kilovatios producidos por hora (Kwh): *D = "C" / 1.43*. Factor de Perdida: 1.43
- Simbolice por colores graduados usando el campo *D* en 3 clases en *Mode = Equal Count (Quantile)* y la paleta Viridis invertida
- Esta pregunta solo será válida si realiza los cálculos solicitados en QGIS

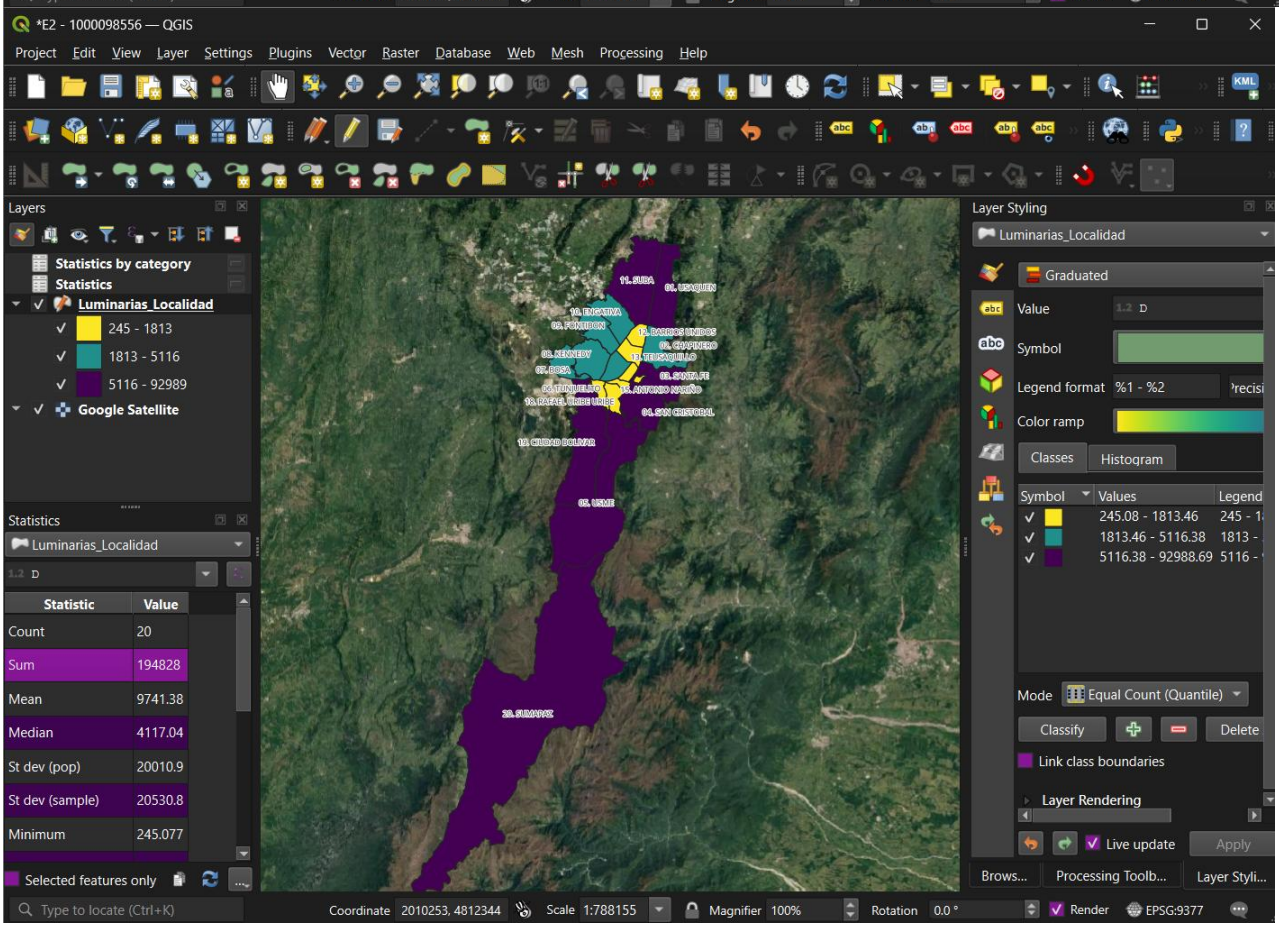


- [x] (A) Promedio (mean) de área usable por localidad (m²): 81657
- [x] (B) Total de páneles solares en Bogotá D.C.: 844253
- [x] (C) Total de potencial de Kilovatios producidos por hora (Kwh): 278603
- [x] (D) Total de real de Kilovatios producidos por hora (Kwh): 194828

Capturas de pantalla:


*E2 - 1000098556 — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Layers
Statistics by category
Statistics
Luminarias_Localidad
245 - 1813
1813 - 5116
5116 - 92989
Google Satellite
Statistics
Luminarias_Localidad
Statistic Value
Count 20
Sum 1.63314e+06
Mean 81657
Median 34511.4
St dev (pop) 167740
St dev (sample) 172097
Minimum 2056.02
Selected features only
Layer Styling
Luminarias_Localidad
Graduated
Value
Symbol
Legend format %1 - %2
Color ramp
Classes Histogram
Symbol Values Legend
245.08 - 1813.46
1813.46 - 5116.38
5116.38 - 92988.69
Mode Equal Count (Quantile)
Classify Delete
Link class boundaries
Layer Rendering
Live update Apply
Brows... Processing Toolb... Layer Styli...
Type to locate (Ctrl+K)
Coordinate 2031106, 4811927
Scale 1:788155
Magnifier 100%
Rotation 0.0 °
Render EPSG:9377
*E2 - 1000098556 — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Layers
Statistics by category
Statistics
Luminarias_Localidad
245 - 1813
1813 - 5116
5116 - 92989
Google Satellite
Statistics
Luminarias_Localidad
Statistic Value
Count 20
Sum 844253
Mean 42212.7
Median 17840.5
St dev (pop) 86714
St dev (sample) 88966.7
Minimum 1062
Selected features only
Layer Styling
Luminarias_Localidad
Graduated
Value
Symbol
Legend format %1 - %2
Color ramp
Classes Histogram
Symbol Values Legend
245.08 - 1813.46
1813.46 - 5116.38
5116.38 - 92988.69
Mode Equal Count (Quantile)
Classify Delete
Link class boundaries
Layer Rendering
Live update Apply
Brows... Processing Toolb... Layer Styli...
Type to locate (Ctrl+K)
Coordinate 2015049, 4811718
Scale 1:788155
Magnifier 100%
Rotation 0.0 °
Render EPSG:9377


*E2 - 1000098556 — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Layers
Statistics by category
Statistics
Luminarias_Localidad
245 - 1813
1813 - 5116
5116 - 92989
Google Satellite
Statistics
Luminarias_Localidad
Statistic Value
Count 20
Sum 278603
Mean 13930.2
Median 5887.36
St dev (pop) 28615.6
St dev (sample) 29359
Minimum 350.46
Selected features only
Layer Styling
Luminarias_Localidad
Graduated
Value
Symbol
Legend format %1 - %2
Color ramp
Classes Histogram
Symbol Values Legend
245.08 - 1813.46
1813.46 - 5116.38
5116.38 - 92988.69
Mode Equal Count (Quantile)
Classify Delete
Link class boundaries
Layer Rendering
Live update Apply
Brows... Processing Toolb... Layer Styli...
Type to locate (Ctrl+K)
Coordinate 2014632, 4813178 Scale 1:788155 Magnifier 100% Rotation 0.0 ° Render EPSG:9377


*E2 - 1000098556 — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Layers
Statistics by category
Statistics
Luminarias_Localidad
245 - 1813
1813 - 5116
5116 - 92989
Google Satellite
Statistics
Luminarias_Localidad
Statistic Value
Count 20
Sum 194828
Mean 9741.38
Median 4117.04
St dev (pop) 20010.9
St dev (sample) 20530.8
Minimum 245.077
Selected features only
Layer Styling
Luminarias_Localidad
Graduated
Value
Symbol
Legend format %1 - %2
Color ramp
Classes Histogram
Symbol Values Legend
245.08 - 1813.46
1813.46 - 5116.38
5116.38 - 92988.69
Mode Equal Count (Quantile)
Classify Delete
Link class boundaries
Layer Rendering
Live update Apply
Brows... Processing Toolb... Layer Styli...
Type to locate (Ctrl+K)
Coordinate 2010253, 4812344 Scale 1:788155 Magnifier 100% Rotation 0.0 ° Render EPSG:9377

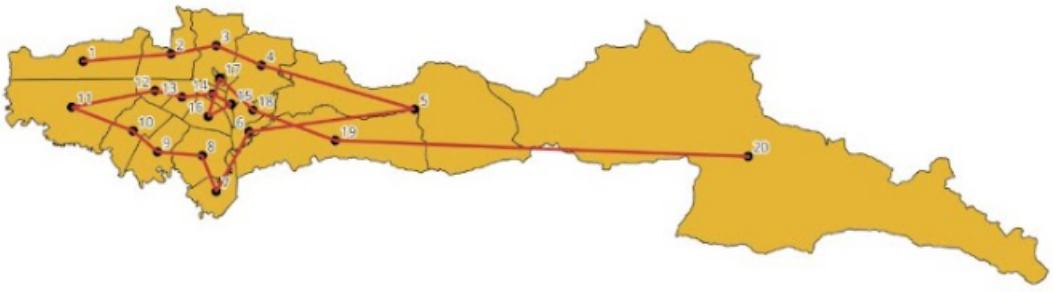
7. P7

Texto de la actividad:

7. Polilinea conectora de centroides de localidades: obtenga las coordenadas CX y CY en metros del centroide de las localidades y cree una polilínea conectora abierta en la secuencia correspondiente al código de la localidad * 4 puntos

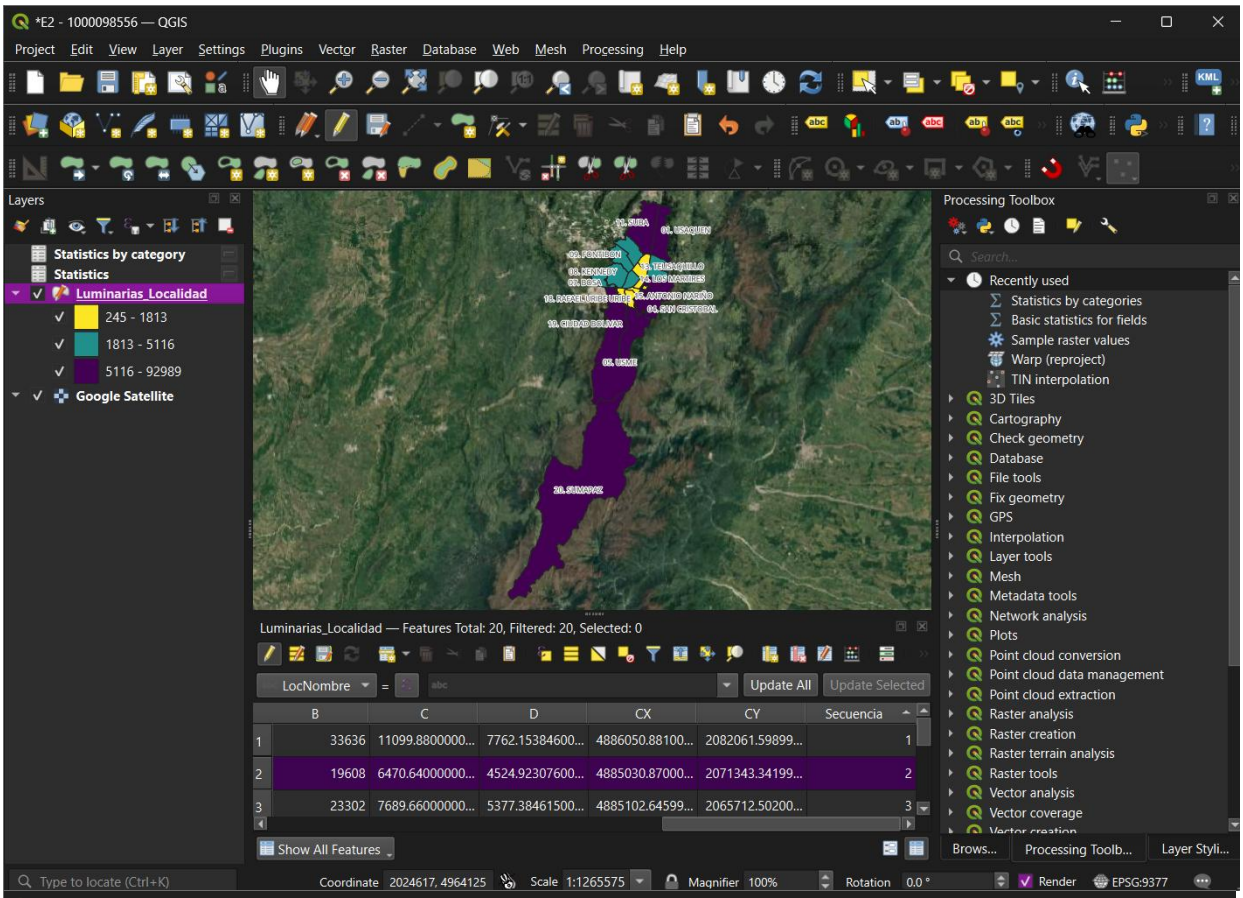
Procedimiento:

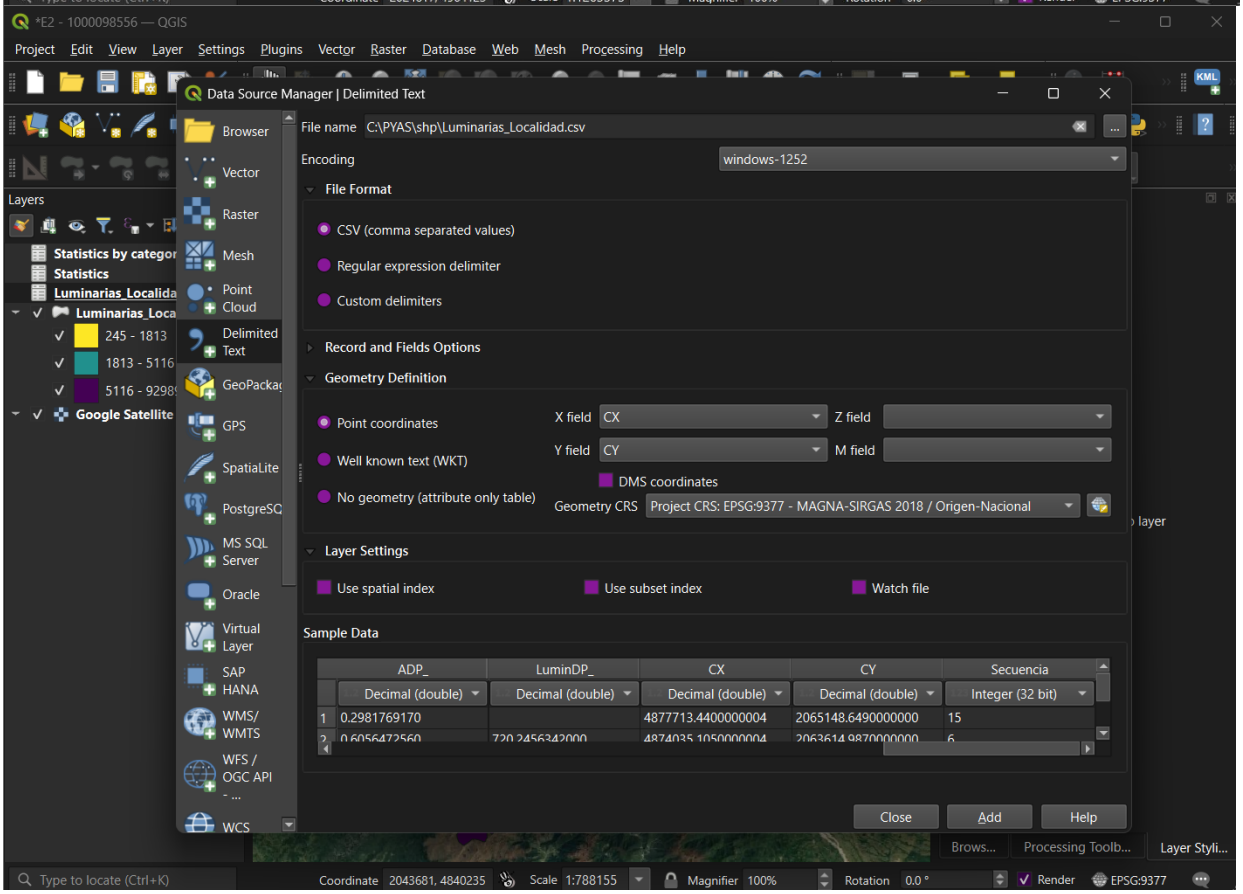
- En la capa *Luminarias_Localidad.shp*, cree los campos tipo real *CX* y *CY* (10 posiciones enteras y 10 decimales de precisión) y desde el *Field Calculator* obtenga sus coordenadas. *CX = x(@geometry), CY = y(@geometry)*
- Cree un campo entero corto con el nombre *Secuencia* y asigne el valor contenido en el campo *LocCodigo*
- Desde el panel *Layers* y dando clic derecho en la capa, exporte la capa *Luminarias_Localidad.shp* a un archivo de texto separado por comas con el nombre *Luminarias_Localidad.csv*
- Desde el menú *Layer / Add Layer / Add Delimited Text Layer...* cree una capa temporal de puntos asignando en *Geometry / Point Coordinates / X field con CX, Y field con CY y Geometry CRS* con el *EPSG: 9377*
- A partir de la capa de nodos y utilizado la herramienta *Vector Creation / Points To Path,* cree la poilínea abierta que conecta los centroides de las localidades en el orden establecido en el campo *Secuencia*
- Desde el *Field Calculator,* calcule la longitud en kilómetros total de la polilínea conectora. *LPkm = length(@geometry)/1000*
- Esta pregunta solo será válida si realiza los cálculos solicitados en QGIS

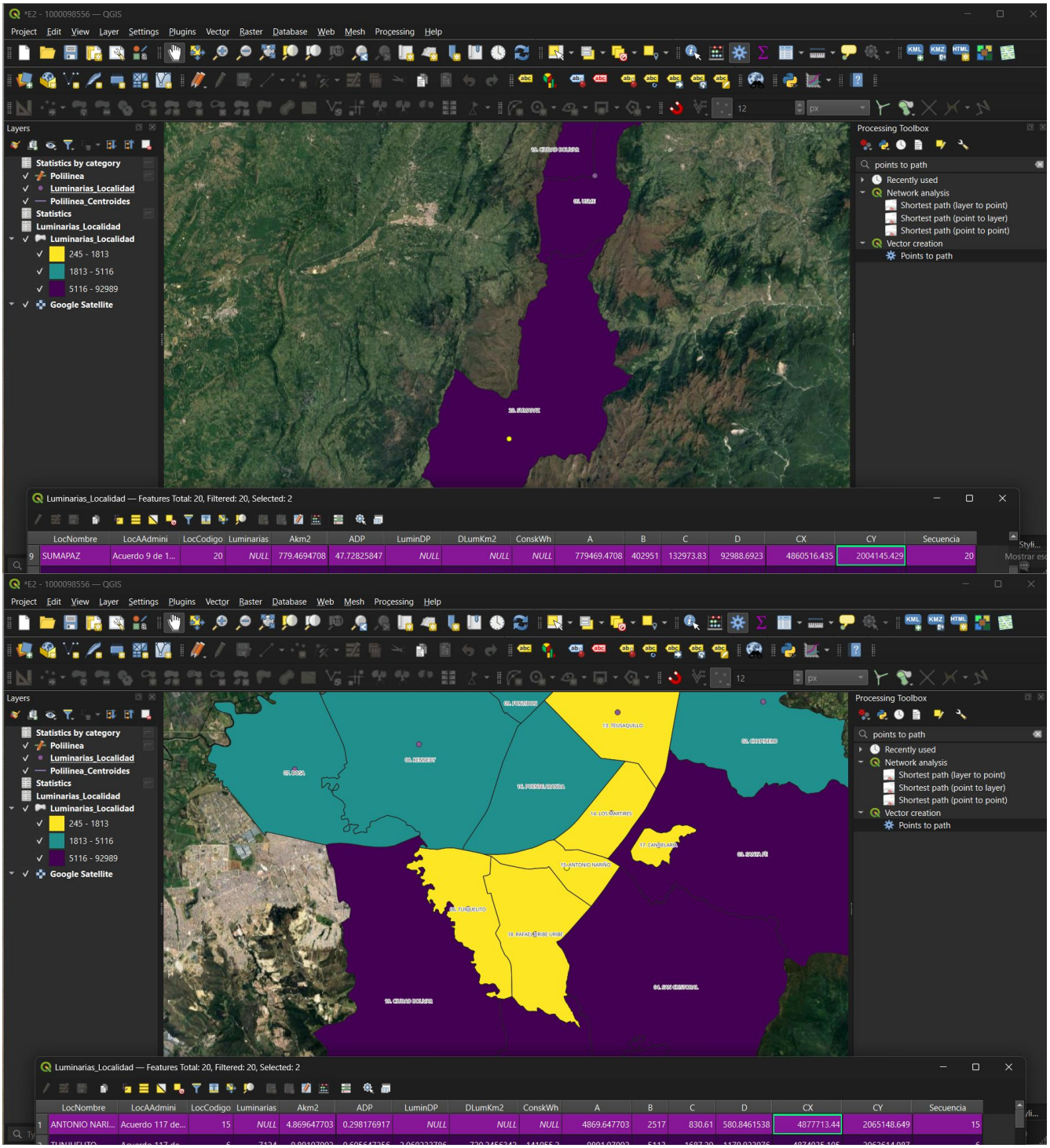


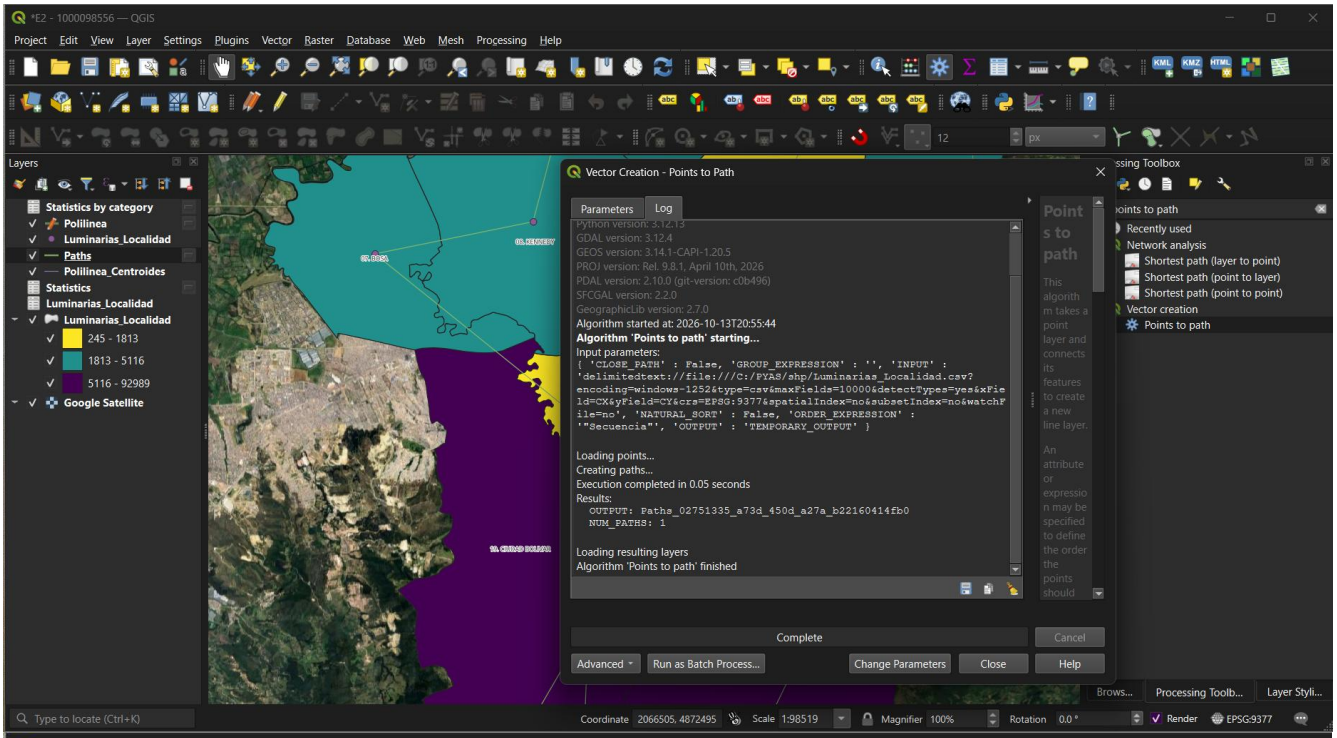
- [x] Longitud total polilínea (km): 187.7759
- [x] # de nodos de la polilínea: 20
- [x] LocCodigo = 15 (Antonio Nariño), CX: 4877713.441
- [x] LocCodigo = 20 (Sumapaz), CY: 2004145.429
- [ ] (No resuelto)

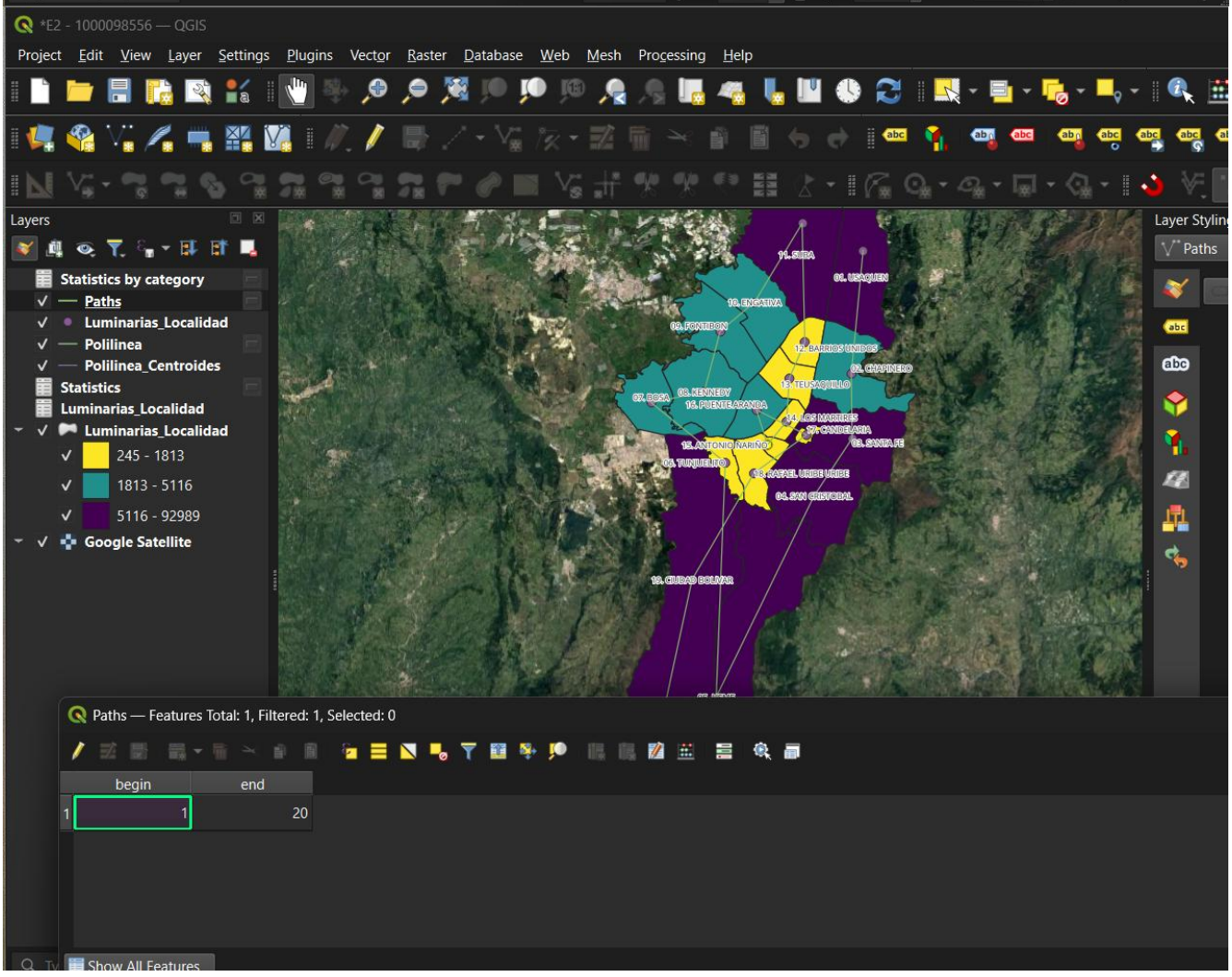
Capturas de pantalla:

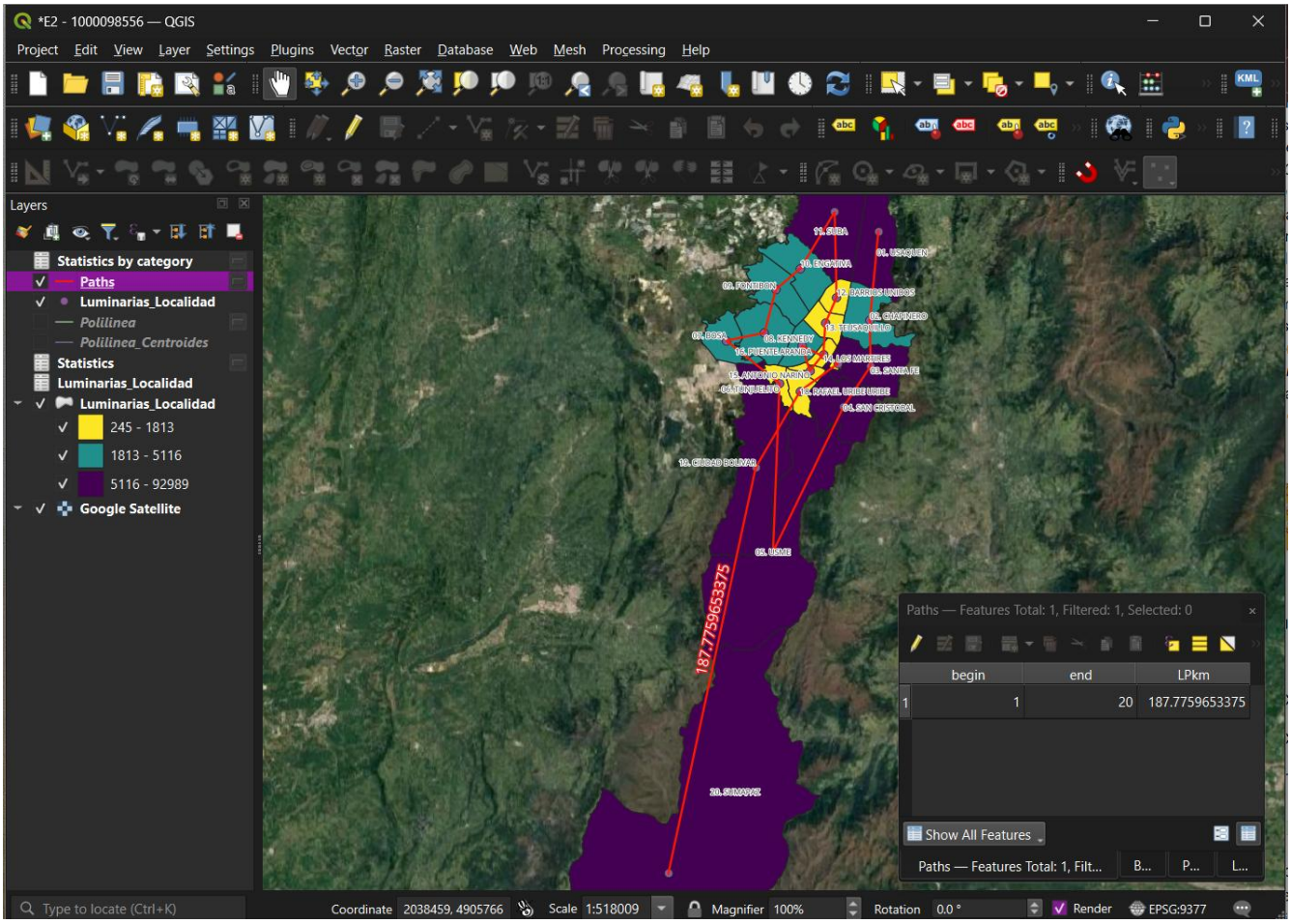





*E2 - 1000098556 — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Layers
Statistics by category
Polilinea
Luminarias_Localidad
Polilinea_Centroides
Statistics
Luminarias_Localidad
Luminarias_Localidad
245 - 1813
1813 - 5116
5116 - 92989
Google Satellite
Processing Toolbox
points to path
Recently used
Network analysis
Shortest path (layer to point)
Shortest path (point to layer)
Shortest path (point to point)
Vector creation
Points to path
Luminarias_Localidad — Features Total: 20, Filtered: 20, Selected: 2
LocNombre LocAAdmini LocCodigo Luminarias Akm2 ADP LuminDP DLumKm2 ConskWh A B C D CX CY Secuencia
9 SUMAPAZ Acuerdo 9 de 1... 20 NULL 779.4694708 47.72825847 NULL NULL NULL 779469.4708 402951 132973.83 92988.6923 4860516.435 2004145.429 20
*E2 - 1000098556 — QGIS
Luminarias_Localidad — Features Total: 20, Filtered: 20, Selected: 2
LocNombre LocAAdmini LocCodigo Luminarias Akm2 ADP LuminDP DLumKm2 ConskWh A B C D CX CY Secuencia
1 ANTONIO NARI... Acuerdo 117 de... 15 NULL 4.869647703 0.298176917 NULL NULL NULL 4869.647703 2517 830.61 580.8461538 4877713.44 2065148.649 15


*E2 - 1000098556 — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Layers
Statistics by category
Polilinea
Luminarias_Localidad
Paths
Polilinea_Centroides
Statistics
Luminarias_Localidad
Luminarias_Localidad
245 - 1813
1813 - 5116
5116 - 92989
Google Satellite
Vector Creation - Points to Path
Parameters
Log
Algorithm started at: 2026-10-13T20:55:44
Algorithm 'Points to path' starting...
Input parameters:
Loading points...
Creating paths...
Execution completed in 0.05 seconds
Results:
OUTPUT: Paths_02751335_a73d_450d_a27a_b22160414fb0
NUM_PATHS: 1
Loading resulting layers
Algorithm 'Points to path' finished
Complete
Cancel
Advanced
Run as Batch Process...
Change Parameters
Close
Help
Processing Toolbox
Points to path
Recently used
Network analysis
Shortest path (layer to point)
Shortest path (point to layer)
Shortest path (point to point)
Vector creation
Points to path


*E2 - 1000098556 — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Layers
Statistics by category
Paths
Luminarias_Localidad
Polilinea
Polilinea_Centroides
Statistics
Luminarias_Localidad
Luminarias_Localidad
245 - 1813
1813 - 5116
5116 - 92989
Google Satellite
Layer Styling
Paths
01. SUBA
02. USAQUEN
12. ENGATIVA
03. FONTIBON
12. BARRIOS UNIDOS
08. CHAPINERO
13. TEUSAQUILLO
07. BOSA
10. KENNEDY
16. PUENTE ARANDA
14. LOS MARTIRES
17. CANDELARIA
15. SANTA FE
19. ANTONIO NARIÑO
05. TUNJUELITO
18. RAFAEL URIBE URIBE
04. SAN CRISTOBAL
19. CIUDAD BOLIVAR
Paths — Features Total: 1, Filtered: 1, Selected: 0
begin
end
1
1
20
Show All Features


*E2 - 1000098556 — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Layers
Statistics by category
Paths
Luminarias_Localidad
Polilinea
Polilinea_Centroides
Statistics
Luminarias_Localidad
Luminarias_Localidad
245 - 1813
1813 - 5116
5116 - 92989
Google Satellite
187.7759653375
Paths — Features Total: 1, Filtered: 1, Selected: 0
begin
end
LPkm
1
1
20
187.7759653375
Show All Features
Paths — Features Total: 1, Filt...
B...
P...
L...
Type to locate (Ctrl+K)
Coordinate 2038459, 4905766
Scale 1:518009
Magnifier 100%
Rotation 0.0 °
Render
EPSG:9377